

# Highlighted Manuscript: Change and Deletion Index

Baseline: 33f5e3d37a85641b13bffa7fc34b94eb5d35c47. Selected as the latest manuscript before August 13 under the author's instruction. Yellow marks show inserted/replaced rendered words; orange frames identify pages containing changed figure assets. The clean PDF text is unchanged by annotations. Source-only layout/preamble edits are preserved in reviewed-baseline-source.diff. This index records removed text that cannot be highlighted in the revised manuscript.

## ***Change 1: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** SPICE

**After:** Structural Circuit

## ***Change 2: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** Automated digitization of

**After:** Recovering a netlist from a

## ***Change 3: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** circuit schematics into simulation-ready SPICE netlists remains challenging because connectivity recovery is brittle under noisy

**After:** schematic requires deciding which

## ***Change 4: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** detections.

**After:** strokes connect which component terminals.

## ***Change 5: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** that converts a scanned schematic into a SPICE netlist through component detection, an

**After:** for this structural connectivity task. An

## ***Change 6: insert | old pp. [] | new pp. [1]***

**Before (including deletions):** (none)

**After:** extractor masks components before binarization and returns

## ***Change 7: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** extractor,

**After:** segments with explicit endpoints. A typed endpoint graph links endpoints, pins and wire bodies under scale-relative tolerances,

## ***Change 8: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** graph-based

**After:** degree-budget completion step reconnects floating pins by bounded assignment. We evaluate on two real- image connectivity benchmarks from CGHD-1152: 31 images with human-verified nets, and 164 held-out photographs from 24 of the corpus's 25 drafters whose reference nets we derive from the dataset's own stroke and symbol annotations. With annotated component boxes, the

## ***Change 9: insert | old pp. [] | new pp. [1]***

**Before (including deletions):** (none)

**After:** reaches component-pair micro-F1 0.890 on the human-verified set, on which its configuration was selected, and 0.711 on the held-out photographs. On both benchmarks it outperforms every deterministic baseline, including connected-component tracing and Hough line fitting, in paired tests with Holm correction. With annotated wires, its score stays within 0.005 of the native value when the images are rescaled from 0.5× to 3×. Replacing the annotated boxes with a trained component detector gives micro-F1 0.627, and most of

### ***Change 10: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** resolves electrical connectivity.

**After:** loss comes from missed components.

### ***Change 11: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** main contribution

**After:** wire extractor reaches F1 0.973 on 134 annotated images; on the held-out photographs, wire extraction

### ***Change 12: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** joining stage: an endpoint-graph model over wire endpoints and component pins with five edge types, combined with degree-budget completion that recovers dropped connections through min- cost b-matching while bounding over-merge with per-pin budgets and self-loop guards. We also introduce a synthetic perturbation protocol and, to our knowledge, the first human-verified net-level connectivity benchmark for hand-drawn circuits, covering 31 images from CGHD-1152. The occlusion-first extractor reaches wire-detection F1 = 0.976 on 134 scanned images. On the human-verified net-level benchmark, the best join reaches connectivity micro-F1 = 0.890, outperforming the prior completion default (0.83), a legacy radius union-find join (0.67), a connected-component net-tracing baseline (0.62), and a Hough- plus-proximity line tracer (0.81), all under identical detected wires. Running the join on perfect wire labels does not improve micro-F1, showing that connectivity, not wire detection, remains the bottleneck.

**After:** main source of error.

### ***Change 13: delete | old pp. [1] | new pp. []***

**Before (including deletions):** first- party comparison shows a strong

**After:** (none)

### ***Change 14: insert | old pp. [] | new pp. [1]***

**Before (including deletions):** (none)

**After:** given the same component boxes

### ***Change 15: insert | old pp. [] | new pp. [1]***

**Before (including deletions):** (none)

**After:** micro-F1

### ***Change 16: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** same images, statistically indistinguishable

**After:** human-verified set: the pooled difference

### ***Change 17: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** but far more expensive

**After:** is not significant,

### ***Change 18: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** without structural guarantees, supporting explicit geometric connectivity recovery for deployable circuit digitization.

**After:** per-image scores favor the model. We release the code, both benchmarks and the evaluation scripts.

### ***Change 19: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** SPICE,

**After:** structural netlist,

### ***Change 20: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** Analog

**After:** A

***Change 21: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** schematics encode functional information (component

**After:** schematic encodes component

***Change 22: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** values,

**After:** component values

***Change 23: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** topological connections)

**After:** the connections between components

***Change 24: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** visual format designed

**After:** drawing made

***Change 25: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** human interpretation. Converting scanned

**After:** people. A netlist records the same connections in a form that software can check, simulate and search. Converting hand-drawn

***Change 26: delete | old pp. [1] | new pp. []***

**Before (including deletions):** machine-readable SPICE (Simulation Program with Integrated Circuit Emphasis)

**After:** (none)

***Change 27: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** enables automated simulation, design verification,

**After:** would let course material, lab notebooks

***Change 28: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** large-scale circuit analysis. While recent advances

**After:** legacy designs enter these tools without manual re-entry. This paper addresses the structural part of that conversion: given the components

***Change 29: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** object detection have made component recognition increasingly reliable, the joining problem, determining which wire segments connect to

**After:** an image, decide

***Change 30: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** pins and to each other, remains the primary failure mode in end-to-end digitization pipelines. The challenge

**After:** terminals share an electrical node. We call this step joining. Joining

***Change 31: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** threefold.

**After:** hard for three reasons.

***Change 32: delete | old pp. [1] | new pp. []***

**Before (including deletions):** inherently

**After:** (none)

***Change 33: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** produces endpoint positions

**After:** yields endpoints at different off-sets from the pins, and strokes stop short of or overshoot their targets. Second, component symbols overlap the wire endpoints

***Change 34: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** vary by 10–80 pixels. Second, components occupy bounding boxes that overlap with wire endpoints, creating ambiguity about whether

**After:** attach to them, so

***Change 35: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** connects

**After:** near a component may belong

***Change 36: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** a component's pin

**After:** one of its pins

***Change 37: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** passes nearby.

**After:** pass by.

***Change 38: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** fragments,

**After:** fragments

***Change 39: replace | old pp. [1] | new pp. [1]***

**Before (including deletions):** crossing wires create complex connectivity patterns that simple radius-based approaches cannot resolve without over-merging.

**After:** wire crossings all look alike to a proximity rule.

***Change 40: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** 88.5%

**After:** 89.0%

***Change 41: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** SPICE

**After:** Structural

***Change 42: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** Pin discovery + simulation

**After:** Inferred pin-to-node map

***Change 43: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** End-to-end pipeline

**After:** Pipeline

***Change 44: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** scanning circuit diagram digitization. Our contributions span the

**After:** structural circuit-netlist recovery. The highlighted stages extract

***Change 45: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** extraction

**After:** endpoints

***Change 46: delete | old pp. [2] | new pp. []***

**Before (including deletions):** joining stages; the

**After:** (none)

***Change 47: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** (dashed red) determines which extracted wire segments connect to which component pins and is the dominant failure mode in end-to-end digitization.

**After:** them into nets.

***Change 48: delete | old pp. [2] | new pp. []***

**Before (including deletions):** real

**After:** (none)

***Change 49: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** circuit images,

**After:** circuits,

***Change 50: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** our

**After:** the

***Change 51: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** (connectivity

**After:** (component-pair

***Change 52: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** (36

**After:** (35

***Change 53: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** 12 nets); right:

**After:** 7 nets). Right:

***Change 54: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** (12

**After:** (11

***Change 55: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** shows, in order: original scan,

**After:** shows the input,

***Change 56: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** close,

**After:** closing,

***Change 57: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** (red overlay),

**After:** (red)

***Change 58: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** every recovered

**After:** each

***Change 59: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** is a distinct colour drawn along

**After:** has

***Change 60: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** conductors; component bounding

**After:** own color;

***Change 61: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** ground-truth labels (as used throughout the net-level evaluation, isolating the join),

**After:** annotated components,

***Change 62: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** mark junctions,

**After:** are junctions

***Change 63: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** mark

**After:** are

***Change 64: insert | old pp. [] | new pp. [2]***

**Before (including deletions):** (none)

**After:** Most image-to-netlist systems erase the components and read nets from the connected components or contours of the remaining wire pixels [1], [2].

***Change 65: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** natural alternative is

**After:** single stroke gap then splits a net, and a single touching stroke merges two. Other systems pass the image

***Change 66: delete | old pp. [2] | new pp. []***

**Before (including deletions):** prompt

**After:** (none)

### ***Change 67: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** vision-language

**After:** language

### ***Change 68: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** (VLM) to

**After:** (LLM) and

### ***Change 69: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** the schematic

**After:** connectivity from its answer [3]. We keep an explicit representation of every wire endpoint

### ***Change 70: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** emit its netlist directly. We test this in Section IV-F: a strong VLM attains connectivity accuracy statistically indistinguishable from our

**After:** decide each connection with

### ***Change 71: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** pipeline, but it does so at orders-of-magnitude higher cost, returns free-form text rather than a structured netlist, offers no structural guarantee against electrically invalid output (e.g. shorted two-terminal parts), and drops connections on the most complex circuits. These properties motivate an explicit, inspectable geometric pipeline whose output is a netlist by construction and whose accuracy matches the VLM's while being deterministic and simulatable. We address these challenges with a complete deterministic pipeline whose novel elements are: (1) an

**After:** rules that can be inspected. This paper makes four contributions: 1) An

### ***Change 72: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** masks detected components

**After:** fills each component region with its local median color

### ***Change 73: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** binarization

**After:** Sauvola binarization

### ***Change 74: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** estimates per-segment endpoints, producing a line-segment-with-endpoints representation (rather than tracing nets directly into connected components, as prior systems do) that is what makes graph-based joining possible; (2) an

**After:** returns line segments with PCA-estimated endpoints. It reaches wire-detection F1 0.973 on 134 CGHD-1152 images (0.976 for the best of 36 variants). 2) A typed

### ***Change 75: delete | old pp. [2] | new pp. []***

**Before (including deletions):** model that constructs a graph over both wire endpoints and component pins,

**After:** (none)

### ***Change 76: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** types capturing distinct

**After:** rules, component-first pin assignment and scale-relative tolerances, followed by degree-budget completion, which reconnects floating pins through a bounded-capacity assignment with a shared-component guard. 3) Two real-image

### ***Change 77: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** patterns; and (3) a degree-budget completion algorithm that identifies and reconnects dropped connections using constrained optimization. We evaluate on the CGHD-1152 dataset [1]

**After:** benchmarks: 31 CGHD- 1152 images

### ***Change 78: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** both a synthetic error injection framework that enables systematic robustness analysis

**After:** human-verified nets,

### ***Change 79: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** human-verified net-level benchmark.

**After:** 164- image held-out benchmark covering 24 of the 25 CGHD drafters whose reference nets are derived from the CGHD authors' own stroke-segmentation maps and symbol polygons. The derived reference agrees with the human nets on 17 overlapping images at micro-F1 0.939, and a human audit of 22 random held-out images found it exact on 21. 4) An evaluation protocol with paired significance tests under Holm correction, complexity and component- size strata, controlled image rescaling, end-to-end runs with a trained component detector, and a vision- language model (VLM) reference that receives the same component boxes as our join. The output is a structural netlist; recognizing component values and device models is outside the scope of this work. Section II reviews related work, Section III describes the method and Section IV the experimental setup. Section V reports results, and Section VI discusses them together with the scope and limitations of the study.

### ***Change 80: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** The end-to-end conversion of

**After:** Several systems convert

### ***Change 81: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** machine-readable netlists has progressed rapidly.

**After:** machine- readable netlists.

### ***Change 82: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** [2]

**After:** [4]

### ***Change 83: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** pattern-recognition

**After:** pattern- recognition

### ***Change 84: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** digitizing circuit

**After:** printed

### ***Change 85: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** combining

**After:** combining

### ***Change 86: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** [3], the closest contemporary system,

**After:** [1]

### ***Change 87: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** connected-component



**After:** connected- component

***Change 88: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** invokes

**After:** uses

***Change 89: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** vision-language model (VLM)

**After:** VLM only

***Change 90: delete | old pp. [2] | new pp. []***

**Before (including deletions):** confines the VLM to symbolic labeling and

**After:** (none)

***Change 91: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** proximity

**After:** proxim- ity

***Change 92: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** trusting

**After:** from

***Change 93: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** VLM with connec- tivity, an insight we share (Section II).

**After:** VLM.

***Change 94: replace | old pp. [2] | new pp. [2]***

**Before (including deletions):** [4] (Auto-SPICE)

**After:** [3]

***Change 95: insert | old pp. [] | new pp. [2]***

**Before (including deletions):** (none)

**After:** 2 VOLUME 11, 2023

***Change 96: replace | old pp. [2] | new pp. [3]***

**Before (including deletions):** produce

**After:** generate

***Change 97: replace | old pp. [2] | new pp. [3]***

**Before (including deletions):** netlists from

**After:** descriptions of

***Change 98: replace | old pp. [2] | new pp. [3]***

**Before (including deletions):** Most directly com- parable to our setting, the

**After:** The

***Change 99: replace | old pp. [2] | new pp. [3]***

**Before (including deletions):** [5] (IEEE Access, 2026)

**After:** [2]

**Change 100: replace | old pp. [2] | new pp. [3]**

**Before (including deletions):** a fully

**After:** an

**Change 101: replace | old pp. [2] | new pp. [3]**

**Before (including deletions):** image-to- SPICE

**After:** image-to-SPICE

**Change 102: replace | old pp. [2] | new pp. [3]**

**Before (including deletions):** circuits, pairing

**After:** circuits. It pairs

**Change 103: replace | old pp. [2] | new pp. [3]**

**Before (including deletions):** module (components

**After:** module, in which components

**Change 104: replace | old pp. [2] | new pp. [3]**

**Before (including deletions):** pixels) and, as we do,

**After:** pixels, and it

**Change 105: delete | old pp. [2] | new pp. []**

**Before (including deletions):** automated

**After:** (none)

**Change 106: replace | old pp. [2, 3] | new pp. [3]**

**Before (including deletions):** simula- 2 VOLUME 11, 2023 tion

**After:** simulation

**Change 107: replace | old pp. [3] | new pp. [3]**

**Before (including deletions):** their

**After:** an

**Change 108: replace | old pp. [3] | new pp. [3]**

**Before (including deletions):** seven-topology

**After:** seven- topology

**Change 109: replace | old pp. [3] | new pp. [3]**

**Before (including deletions):** they report

**After:** it reports

**Change 110: replace | old pp. [3] | new pp. [3]**

**Before (including deletions):** Their

**After:** Its

**Change 111: replace | old pp. [3] | new pp. [3]**

**Before (including deletions):** detection is an instance of

**After:** detection belongs to

***Change 112: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** connected-component/contour recipe

**After:** connected-component family that

***Change 113: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** evaluate

**After:** evaluate

***Change 114: insert | old pp. [] | new pp. [3]***

**Before (including deletions):** (none)

**After:** on identical wire inputs

***Change 115: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** IV-D); because their

**After:** V-C); its

***Change 116: delete | old pp. [3] | new pp. []***

**Before (including deletions):** (transistor-level analog, authored in-house)

**After:** (none)

***Change 117: delete | old pp. [3] | new pp. []***

**Before (including deletions):** (LT- spice node-voltage pass/fail)

**After:** (none)

***Change 118: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** ours (discrete components on CGHD-1152, component-pair F1),

**After:** ours, so

***Change 119: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** comparable, but the comparison shows contour/CCL node detection remains the prevailing approach, which our endpoint-graph join substantially outperforms on a controlled, identical-input benchmark. Instance-segmentation approaches extract circuit graphs directly from pixels:

**After:** comparable.

***Change 120: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** [6]

**After:** [5]

***Change 121: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** components

**After:** components

***Change 122: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** over

**After:** on

***Change 123: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** in- puts,

**After:** inputs,

***Change 124: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** [7]

**After:** [6]

***Change 125: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** recent JUHCCR- v1

**After:** JUHCCR-v1

***Change 126: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** [8]

**After:** [7]

***Change 127: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** general-purpose object

**After:** general- purpose

***Change 128: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** YOLO

**After:** You Only Look Once (YOLO)

***Change 129: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** [9], [10]

**After:** [8], [9]

***Change 130: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** [11], with the ultimate goal of producing simulator-ready descriptions in

**After:** [10], and target simulator input

***Change 131: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** [12]. These works establish detection

**After:** [11]. SPICE- level modeling also supports hardware work in neighbor- ing domains, including memristor-based neuromorphic cir- cuits that realize brain-inspired learning [12]

***Change 132: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** recognition as largely solved;

**After:** memristive decision-making circuits for autonomous navigation [13]. We focus on

***Change 133: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** bottleneck shifts to reconstructing electrical connectivity, the focus of our contribution.

**After:** joining stage and evaluate it with both annotated and detected components.

***Change 134: delete | old pp. [3] | new pp. []***

**Before (including deletions):** hand- drawing

**After:** (none)

***Change 135: replace | old pp. [3] | new pp. [3]***

Before (including deletions): [1]

After: [14]

***Change 136: replace | old pp. [3] | new pp. [3]***

Before (including deletions): an- notations

After: annota- tions

***Change 137: replace | old pp. [3] | new pp. [3]***

Before (including deletions): analog/mixed- signal

After: analog/mixed-signal

***Change 138: replace | old pp. [3] | new pp. [3]***

Before (including deletions): [13]

After: [15]

***Change 139: replace | old pp. [3] | new pp. [3]***

Before (including deletions): detec- tion, while

After: detection, and

***Change 140: replace | old pp. [3] | new pp. [3]***

Before (including deletions): [14] is a benchmark for evaluat- ing

After: [16] evaluates

***Change 141: replace | old pp. [3] | new pp. [3]***

Before (including deletions): LLM capabilities

After: LLMs

***Change 142: delete | old pp. [3] | new pp. []***

Before (including deletions): Beyond circuits,

After: (none)

***Change 143: replace | old pp. [3] | new pp. [3]***

Before (including deletions): [15]

After: [17]

***Change 144: replace | old pp. [3] | new pp. [3]***

Before (including deletions): an end-to-end

After: a

***Change 145: replace | old pp. [3] | new pp. [3]***

Before (including deletions): di- agrams,

After: diagrams,

***Change 146: replace | old pp. [3] | new pp. [3]***

Before (including deletions): an explicit

After: a

***Change 147: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** We use these benchmarks both as evidence of where current systems succeed (component recognition) and where they fail (connectivity), and contribute a synthetic robustness protocol that isolates the join step under controlled detector noise.

**After:** To our knowledge, no public benchmark reports net-level connectivity on hand- drawn circuit images; Peker et al. [2] report whole-netlist success on an in-house hand-drawn set.

***Change 148: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** [16],

**After:** [18]

***Change 149: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** understand- ing in particular.

**After:** understanding.

***Change 150: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** [17]

**After:** [19]

***Change 151: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** using

**After:** with

***Change 152: delete | old pp. [3] | new pp. []***

**Before (including deletions):** (VQA)

**After:** (none)

***Change 153: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** connection- matrix

**After:** connection-matrix

***Change 154: delete | old pp. [3] | new pp. []***

**Before (including deletions):** However, multiple independent bench- marks show that general VLMs handle component recogni- tion far better than connectivity.

**After:** (none)

***Change 155: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** con- nection

**After:** connection

***Change 156: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** [15], leading

**After:** [17],

***Change 157: delete | old pp. [3] | new pp. []***

**Before (including deletions):** very

**After:** (none)

***Change 158: delete | old pp. [3] | new pp. []***

**Before (including deletions):** at

**After:** (none)

***Change 159: delete | old pp. [3] | new pp. []***

**Before (including deletions):** at

**After:** (none)

***Change 160: delete | old pp. [3] | new pp. []***

**Before (including deletions):** and

**After:** (none)

***Change 161: delete | old pp. [3] | new pp. []***

**Before (including deletions):** at

**After:** (none)

***Change 162: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** [14]

**After:** [16]

***Change 163: delete | old pp. [3] | new pp. []***

**Before (including deletions):** MLLMs on

**After:** (none)

***Change 164: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** This corroborates the design choice, shared

**After:** These results vary

***Change 165: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** SINA [3], to keep wire-to-component

**After:** task and input, which motivates measuring VLM

***Change 166: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** out of the VLM and instead solve it with an explicit geometric model. We make this separation concrete and quantify it with

**After:** directly on

***Change 167: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** own Claude-VLM connectivity experiment.

**After:** benchmark (Section V-I).

***Change 168: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** morphological operations,

**After:** morphological operations

***Change 169: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** [18], typically preceded by

**After:** [20], usually after

***Change 170: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** threshold- ing [19]

**After:** thresholding [21]

***Change 171: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** [20].

**After:** [22].

***Change 172: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** dom- inant

**After:** legacy

***Change 173: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** existing pipelines

**After:** our pipeline

***Change 174: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** grabs all

**After:** takes every

***Change 175: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** pins

**After:** pin

***Change 176: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** (typically 30px), connected via transitive union- find [21]. This over-merges when multiple

**After:** (30 px), and connections are merged transitively with union-find [23]. When several

***Change 177: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** shorting

**After:** this rule assigns both pins of a

***Change 178: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** parts (R, C, L, D) onto

**After:** part to

***Change 179: delete | old pp. [3] | new pp. []***

**Before (including deletions):** SINA [3] likewise derives nets from CCL proximity, which is similarly vulnerable to over-merge under detector noise.

**After:** (none)

***Change 180: delete | old pp. [3] | new pp. []***

**Before (including deletions):** instead

**After:** (none)

***Change 181: insert | old pp. [] | new pp. [3]***

**Before (including deletions):** (none)

**After:** instead



***Change 182: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** incorporates

**After:** uses

***Change 183: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** geometry

**After:** geometry,

***Change 184: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** directional preference, and

**After:** we

***Change 185: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** a degree- budget completion:

**After:** degree-budget completion,

***Change 186: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** formulation [22], [23]

**After:** formula- tion [24], [25]

***Change 187: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** per-pin degree and thereby limits over-merge when endpoints are dropped or displaced. This differs from graph

**After:** the number of recovery connections per pin. Graph

***Change 188: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** engineering [24], which

**After:** engineer- ing [26]

***Change 189: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** gate-level

**After:** gate- level

***Change 190: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** a purely geometric model whose

**After:** geometric, and its

***Change 191: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** relationships

**After:** relations

***Change 192: delete | old pp. [3] | new pp. []***

**Before (including deletions):** wire

**After:** (none)

***Change 193: delete | old pp. [3] | new pp. []***

**Before (including deletions):** com- ponent

**After:** (none)

***Change 194: delete | old pp. [3] | new pp. []***

**Before (including deletions):** A complementary recent direction replaces rule- based connectivity with a learned model:

**After:** (none)

***Change 195: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** [25] pre- dicts

**After:** [27] predicts

***Change 196: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** schematics. Such

**After:** schematics; such learned

***Change 197: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** are powerful but require task-specific connec- tivity training data and, trained on printed schematics, do not transfer to hand-drawn images without retraining; our geometric model needs no

**After:** need

***Change 198: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** and is the focus here.

**After:** data, whereas our join needs none.

***Change 199: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** CircuitNet [26] likewise rely on

**After:** Cir- cuitNet [28] combine

***Change 200: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** plus

**After:** with

***Change 201: delete | old pp. [3] | new pp. []***

**Before (including deletions):** six-stage pipeline. The full pipeline proceeds through

**After:** (none)

***Change 202: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** detection via oriented bounding box

**After:** detec- tion with an oriented-bounding-box

***Change 203: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** occlusion (filling detected regions

**After:** occlusion, which fills each component region, enlarged by 15% of its box size (at least 5 px),

***Change 204: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** local

**After:** the

***Change 205: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** color);

**After:** color of its surroundings;

***Change 206: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** adaptive binarization;

**After:** binarization ( $k = 0.285$ , window 67 px) inside the union of component boxes plus 10 px padding;

***Change 207: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** connected component

**After:** connected-component

***Change 208: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** with morphological post-processing;

**After:** after a  $3 \times 3$  elliptical closing, discarding components under 28 px;

***Change 209: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** wire end- point extraction via

**After:** wire-segment and endpoint estimation by

***Change 210: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** on

**After:** of each connected component, followed by overlap deduplication (12■, 8 px) and an anchor filter with a 16 px

***Change 211: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** regions;

**After:** distance;

***Change 212: delete | old pp. [3] | new pp. []***

**Before (including deletions):** wire

**After:** (none)

***Change 213: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** via

**After:** with

***Change 214: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** endpoint-graph model with

**After:** endpoint graph and

***Change 215: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** is a netlist mapping

**After:** maps each

***Change 216: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** pins

**After:** pin

***Change 217: insert | old pp. [] | new pp. [3]***

**Before (including deletions):** (none)

**After:** an

***Change 218: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** nodes, suitable for SPICE netlist generation.

**After:** node.

***Change 219: delete | old pp. [3] | new pp. []***

**Before (including deletions):** representative

**After:** (none)

***Change 220: delete | old pp. [3] | new pp. []***

**Before (including deletions):** a set of

**After:** (none)

***Change 221: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** where

**After:** with

***Change 222: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):**  $\cup P$ .

**After:**  $\cup P$  and five edge types.

***Change 223: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** with weight 0. This ensures

**After:** so that

***Change 224: replace | old pp. [3] | new pp. [3]***

**Before (including deletions):** of a wire are always in

**After:** belong to

***Change 225: delete | old pp. [3] | new pp. []***

**Before (including deletions):** VOLUME 11, 2023 3

**After:** (none)

***Change 226: replace | old pp. [4] | new pp. [3]***

**Before (including deletions):** endpoint pairs (ei, ej)

**After:** endpoints ei, ej on different wires

***Change 227: replace | old pp. [4] | new pp. [3]***

**Before (including deletions):** edge with weight  $\mathbf{e_i - e_j}$ .

**After:** an edge.

***Change 228: replace | old pp. [4] | new pp. [3]***

**Before (including deletions):** captures wire

**After:** rule connects

***Change 229: replace | old pp. [4] | new pp. [3]***

**Before (including deletions):** corners,

**After:** corners

### **Change 230: replace | old pp. [4] | new pp. [3]**

**Before (including deletions):** For each

**After:** Each

### **Change 231: replace | old pp. [4] | new pp. [3]**

**Before (including deletions):** e

**After:** connects to at most one pin. The component-first assignment described below chooses the pin. If no component is assigned, the endpoint takes the pin within  $d \leq \tau_{\text{pin}}$  with the smallest score, where the score is  $d[1 - \alpha \max(0, \cos \theta)]$  with  $\alpha = 0.35$

### **Change 232: delete | old pp. [4] | new pp. []**

**Before (including deletions):** pin p with  $\|e - p\| \leq \tau_{\text{pin}}$ , add edge with weight  $\|e - p\| \cdot (1 - \alpha \cdot \cos \theta)$ , where

**After:** (none)

### **Change 233: delete | old pp. [4] | new pp. []**

**Before (including deletions):** is

**After:** (none)

### **Change 234: insert | old pp. [] | new pp. [3]**

**Before (including deletions):** (none)

**After:** outward

### **Change 235: replace | old pp. [4] | new pp. [3]**

**Before (including deletions):** vector, and  $\alpha$  controls directional preference. This

**After:** vector (for  $d \leq 10 - 6 \text{ px}$  the score

### **Change 236: replace | old pp. [4] | new pp. [3]**

**Before (including deletions):** the critical

**After:** d). If neither step finds a pin, no

### **Change 237: delete | old pp. [4] | new pp. []**

**Before (including deletions):** type: it connects wires to components. Edge type 4 (endpoint–wire–body): For each endpoint e and wire segment  $w_j = (c, d)$ , if the point-to-segment distance  $\text{dist}(e, w_j) \leq \tau$ , add edge with weight  $\text{dist}(e, w_j)$ . This captures T-junctions where a wire terminates on another wire's body. Edge type 5 (pin–wire–body): For each component pin p and wire segment  $w_j$ , if the point-to-segment distance  $\text{dist}(p, w_j) \leq \tau$  and the projection lands mid-span (not at  $w_j$ 's endpoints, which are handled by edge type 3), add edge with weight  $\text{dist}(p, w_j)$ . This binds components that tap onto a passing rail along the wire's body rather than at its end. It recovers connectivity that endpoint-only binding misses, without the all-to-all over-merge of radius grabbing. The tolerances  $\tau_{\text{join}}$ ,  $\tau_{\text{pin}}$ , and  $\tau$  are scale-relative:  $\tau = k \cdot s$  where s

**After:** (none)

### **Change 238: replace | old pp. [4] | new pp. [3]**

**Before (including deletions):** the characteristic scale (median component diagonal), so a single parameter setting works across circuits of varying size. Component-first pin assignment: For edge type 3, we use a two-step assignment. First, assign each endpoint to its nearest component by bounding-box distance (0 if inside the bbox, using a generous radius of  $\max(\tau_{\text{pin}}, 0.5 \cdot \text{diagonal})$ ). Second, within the assigned component, select the pin by geometric position: for horizontal components ( $w > h$ ), left pin if endpoint  $x < \text{center } x$ , else right pin; for vertical components ( $h > w$ ), top pin if endpoint  $y < \text{center } y$ , else bottom pin. This handles the common case where endpoints land inside a component's bounding box but far from its pin. Figure 3 contrasts the legacy radius join with our endpoint-graph model. Nets are computed as connected components of G, projected onto pins. The result is a netlist mapping each pin to a net identifier. C. DEGREE-BUDGET COMPLETION The endpoint-graph join may leave some component pins “floating”: touching only their own component's net, with no wire connecting them to another component. This indicates a dropped or over-displaced wire endpoint that the detector failed to place near the correct pin. Degree-budget completion identifies these floating pins and reconnects them via constrained

optimization. A pin  $p$  is treated as floating if its net contains only pins from  $p$ 's own component, excluding inert types such as gnd, vss, text, antenna, junction, and terminal. For each floating pin  $p$ , we generate candidate edges to all pins  $q$  in other components within reach  $r = \rho \cdot s$ , where  $\rho$  is the reach factor; we use  $\rho = 4$  for the deployed scale-relative join, and Section IV-D shows a broad F1 plateau for  $\rho \in [3, 5]$ . Candidate edge weight is Euclidean distance  $\|p - q\|$ . We then solve a min-cost b-matching problem,  $\min \sum_{(p,q) \in E'} w(p,q) \cdot x_{pq}$  subject to  $\sum_q x_{pq} \leq 1$  for each floating pin  $p$ , using the Hungarian algorithm [22] (scipy.optimize.linear\_sum\_assignment). Each target pin offers up to three slots so legitimate multi-terminal nets are not starved, and a per-pin opt-out leaves a pin with no good target unconnected rather than forcing a spurious edge. Finally, we reject any matched edge  $(p, q)$  if  $p$  and  $q$  belong to the same component or if adding the edge would merge two nets that already share a component, which structurally prevents short-circuiting two-terminal parts. Completion edges are “wireless”: they represent inferred connections without corresponding wire segments, analogous to manual pin merges in schematic editors. Figure 4 illustrates the completion step.

#### IV. RESULTS

##### A. SYNTHETIC EVALUATION FRAMEWORK

We authored 15 synthetic circuits spanning 3–6 components and 2–6 nets, including parallel resistors, series dividers, Wheatstone bridges, RC/RL circuits, diode loops, and a 6-component ring (cyclic and bridge topologies being the main stress cases for join robustness). Each circuit has a known ground-truth netlist and SPICE deck with a test voltage source. We inject five categories of error at four severity levels (L1–L4) on top of the clean (L0) baseline. These are Gaussian endpoint jitter with standard deviation  $\sigma$  (from 3 px at L1 to 14 px at L4), cut-short corruption in which each wire is shortened inward by a fixed length to model endpoints that stop short of the pin (from 4 px at L1 to 22 px at L4), full wire drops with probability  $p_{\text{drop}}$  (up to 0.20 at L4), wrong-pin snaps with probability  $p_{\text{wp}}$  (from 0.02 at L2 to 0.10 at L4), and anchor displacement, where with probability  $p_{\text{disp}}$  one endpoint is moved by up to  $a_{\text{max}}$  pixels (from 30 px at L1 to 80 px at L4). We report join F1 as the component-connectivity F1 score, measuring which pairs of components share a net in the predicted versus ground-truth netlist. We report simulation accuracy as the percentage of random seeds for which every source current in the predicted SPICE netlist matches the clean oracle within 5%.

##### B. WIRE DETECTION BENCHMARK

We evaluate on 134 images from CGHD-1152 with ground-truth wire and component labels. Figure 5 and Table 1 summarize the top configurations across 36 tested variants. The pipeline achieves median F1 = 1.000 across the benchmark set, with 87% of images scoring F1  $\geq 0.90$ . Sauvola adaptive binarization dominates all alternatives: Otsu yields F1 = 0.789, adaptive Gaussian F1 = 0.845. Component occlusion matters: without it, component internals (text, symbols, internal wiring) generate hundreds of false wire detections.

**After:** added.

#### ***Change 239: insert | old pp. [] | new pp. [3]***

**Before (including deletions):** (none)

**After:** 3

#### ***Change 240: replace | old pp. [5] | new pp. [4]***

**Before (including deletions):** approaches compared.

**After:** approaches.

#### ***Change 241: replace | old pp. [5] | new pp. [4]***

**Before (including deletions):** grabs

**After:** takes

#### ***Change 242: replace | old pp. [5] | new pp. [4]***

**Before (including deletions):** causing over-merging.

**After:** which over-merges.

#### ***Change 243: replace | old pp. [5] | new pp. [4]***

**Before (including deletions):** Our endpoint-graph

**After:** Endpoint-graph

#### ***Change 244: replace | old pp. [5] | new pp. [4]***

**Before (including deletions):** (wire body, endpoint-pin)

**After:** and component-first pin assignment; directional scoring applies only to the fallback pin search. Edge type 4 (endpoint-wire-body): For an endpoint  $e$  and a different segment  $w_j$ , add an edge if the point-to-segment distance is at most  $\tau_t$  and the normalized projection parameter lies in  $[0.15, 0.85]$ . This rule captures T-junctions away from segment ends. Edge

type 5 (pin–wire-body): For a pin  $p$  and a segment  $w_j$ , add an edge if the point-to-segment distance is at most  $\tau$  and the projection parameter lies in  $[0.05, 0.95]$ . This rule captures pins tapped onto a rail. The tolerances are scale-relative

### **Change 245: replace | old pp. [5] | new pp. [4]**

**Before (including deletions):** directional preference resolve connectivity without over-merging.

**After:** fixed pixel clamps:  $\tau_{\text{pin}} = \min(60, \max(24, 0.62s))$ ,  $\tau_{\text{join}} = \min(28, \max(11, 0.30s))$  and  $\tau = \min(20, \max(8, 0.20s))$ , in pixels. Here  $s$  is the median component-box diagonal of the image (the upper middle value for an even count), with a wire-length scale as fallback when component geometry is unavailable. The multipliers and clamps are set by hand. Because  $s$  is one scalar per image, these three tolerances do not vary with component size inside a drawing. Component-first pin assignment: For edge type 3, each endpoint is first assigned to a component. Its distance to a component is 0 inside the component's OBB and otherwise the distance to the nearest OBB edge (axis-aligned box distance when OBB vertices are unavailable). The endpoint is assigned to the nearest component within  $\max(\tau_{\text{pin}}, 0.5 \delta_c)$ , where  $\delta_c$  is that component's own box diagonal. Because this radius grows with each component's diagonal, endpoint-to-component binding follows the size of each component in addition to the image-level scale  $s$ . Within the assigned component, the endpoint takes the nearest pin. Figure 3 contrasts the legacy radius join with the endpoint graph. Nets are the connected components of  $G$ , accumulated with union-find and projected onto pins. The result maps each pin to a net identifier. C. DEGREE-BUDGET COMPLETION After base joining, some pins lie in a net that contains only pins of their own component. Completion treats these floating pins as candidates for recovery, excluding inert types (ground, supply markers, text, antennas, probes, junctions and terminals). Completion keeps the base nets and proposes connections from each floating pin  $p$  to pins  $q$  of other components in other nets. A candidate's distance must not exceed the reach  $r =$

### **Change 246: replace | old pp. [5] | new pp. [4]**

**Before (including deletions):** After

**After:** after

### **Change 247: replace | old pp. [5] | new pp. [4]**

**Before (including deletions):** “floating” (touching)

**After:** floating (connected)

### **Change 248: insert | old pp. [] | new pp. [4]**

**Before (including deletions):** (none)

**After:** to

### **Change 249: replace | old pp. [5] | new pp. [4]**

**Before (including deletions):** Completion

**After:** completion

### **Change 250: replace | old pp. [5] | new pp. [4]**

**Before (including deletions):** via

**After:** by

### **Change 251: replace | old pp. [5] | new pp. [4]**

**Before (including deletions):** dashed),

**After:** dashed); a guard rejects merges between nets that already share a component.  $\rho_{\text{tc}}$ , where  $\tau_c = \min(60, \max(24, 0.62s))$   $p_x$  and  $p = 4$ . For each candidate we look for a wire witness: a detected segment whose endpoints lie near  $p$  and  $q$  in either orientation. The witness cost is the smallest, over segments and orientations, of the larger endpoint-to-pin distance, bounded by  $r$ . Without a witness, the candidate cost is  $\|p - q\|$ . The candidates form a rectangular assignment matrix

### **Change 252: replace | old pp. [5] | new pp. [4]**

**Before (including deletions):** self-loop

**After:** one row per floating pin. Each target pin supplies up to three slot columns, limited by its candidate demand, and each row has a private dummy column of cost  $1.5r$  for opting out. Unavailable assignments cost 106. The Hungarian algorithm [24] (`scipy.optimize.linear_sum_assignment`) assigns every row to one column, with at most one row per column; the target slots

implement bounded-capacity matching [25]. Matches are applied in sequence. Dummy and unavailable assignments add no edge. A merge is skipped if both pins are already in the same net or if their nets already share a component; this

### ***Change 253: replace | old pp. [5] | new pp. [4, 5]***

**Before (including deletions):** preventing shorts

**After:** acts on completion merges only. Completion edges have no corresponding wire segment, much like a manual pin merge in a schematic editor. Figure 4 illustrates the step.

**IV. EXPERIMENTAL SETUP**

**A. DATASETS AND IMAGE PROVENANCE** All real images come from CGHD-1152 [14], a public corpus of hand-drawn circuit images organized by drafter. We use four evaluation sets. The wire benchmark has 134 images with human wire labels and annotated component boxes. The human-verified net benchmark has 31 images from 10 drafters. The 34 annotated images were chosen by circuit content, not appearance: ten initial images spanning a range of complexity, and the rest sampled at even intervals of electrical-component count from the circuits whose parts all belong to the scored device types. Image background was not a selection criterion, but the selected copies have mostly plain backgrounds, partly because 16 of them are stroke maps (see below). Three annotated images were excluded before scoring (one unverified, two failing physical-sanity checks). Net proposals were bootstrapped by a conservative endpoint-graph join over the annotated wires and corrected by one annotator in a purpose-built verification tool. The 31 images contain 3–14 electrical components each (median 7); 15 have 4 VOLUME 11, 2023 at most five and 12 have at least ten. The independent held-out benchmark (164 images, 24 drafters) and the synthetic suite (15 authored circuits) are described in Sections V-D and V-A. The 134 wire-benchmark and 31 net-benchmark images are 704 × 704 copies from a Roboflow export of CGHD. The export resized each original without preserving its aspect ratio. Of the 31 net-benchmark images, 14 are also flipped or rotated relative to the CGHD file, and 16 (45 of the 134 wire-benchmark images) are CGHD’s binary stroke-segmentation renderings instead of photographs. Wire and net labels were drawn on these copies. The held-out benchmark uses the original CGHD photographs.

**B. CONNECTIVITY METRIC** For each image we form the set of unordered pairs of electrical components that share at least one net, and compare predicted with reference pairs. The electrical subset contains resistors, capacitors, inductors, diodes, transistors, sources and integrated circuits; junctions, terminals, ground symbols and crossovers affect the score only through the connections they mediate. The metric ignores which terminal of a component carries a connection. Micro-F1 pools true positives (TP), false positives (FP) and false negatives (FN) over images, so a dense board contributes in proportion to its connections; it is our primary metric. Macro-F1, the mean of per-image F1, is reported alongside.

**C. STATISTICAL PROTOCOL** All comparisons are paired over images. We report 95% percentile confidence intervals (CIs) from an image-level bootstrap with B = 10,000 resamples, and test each difference with three procedures: the bootstrap itself, a sign-flip permutation test that swaps the two methods’ outputs

### ***Change 254: replace | old pp. [5] | new pp. [5]***

**Before (including deletions):** two-terminal components. 0.6 0.7 0.8 0.9 1.0 Wire detection F1 (134 images) Triangle Otsu adaptive

**After:** each image (exact when at most 22 images differ, otherwise 100,000 Monte Carlo draws), and a Wilcoxon signed-rank test on per-image F1. The first two test the micro-F1 difference; the Wilcoxon test compares images. Holm’s correction is applied within each procedure across the comparisons of a table. We also count per-image wins, ties and losses (W/T/L). Unless stated otherwise (Section V-G), every join receives annotated component boxes and the pipeline’s own detected wires.

**V. RESULTS**

**A. SYNTHETIC EVALUATION** We authored 15 circuits with 3–6 components and 2–6 nets, including parallel resistors, series dividers, Wheatstone bridges, RC/RL circuits, diode loops and a 6-component ring. Each circuit has a known netlist, component values and a test voltage source. We inject five error types at four severity levels (L1–L4) on top of the clean level L0:

### ***Change 255: replace | old pp. [5] | new pp. [5]***

**Before (including deletions):** Sauvola (alt.) Sauvola (k=0.2875) Sauvola (w=61) Sauvola,

**After:** endpoint jitter ( $\sigma$  from 3 px at L1 to 14 px at L4); cut-short wires, shortened inward by 4 px at L1 to 22 px at L4; wire drops with probability up to 0.20 at L4; wrong-pin snaps with probability from 0.02 at L2 to 0.10 at L4; and

### ***Change 256: replace | old pp. [5] | new pp. [5]***

**Before (including deletions):** 12px Sauvola, anchor 16px (ours) 0.758 0.789 0.928 0.949 0.950 0.959 0.973 0.976 FIGURE 5. Wire detection F1 across thresholding methods

**After:** displacement, which moves one endpoint by up to 30 px at L1

### ***Change 257: replace | old pp. [5] | new pp. [5]***

**Before (including deletions):** pipeline variants (134 images). Our configuration (Sauvola binarization with a 16



**After:** 80

### ***Change 258: replace | old pp. [5] | new pp. [5]***

**Before (including deletions):** anchor filter) achieves F1 = 0.976. Otsu and triangle thresholding degrade significantly.

**After:** at L4.

### ***Change 259: replace | old pp. [5] | new pp. [5]***

**Before (including deletions):** Wire Detection Performance (134 images) Configuration

**After:** Synthetic join leaderboard: mean component-pair

### ***Change 260: replace | old pp. [5] | new pp. [5]***

**Before (including deletions):** Prec Rec Sauvola + anchor 16 px (ours) 0.976 0.973 0.978 Sauvola + anchor 12 px 0.973 0.974 0.972 Sauvola (w=61) + anchor 12 px 0.959 0.944 0.974 Sauvola (k=0.2875) + anchor 14 px 0.950 0.921 0.980 Otsu threshold + CCL 0.789 0.796 0.783 C. JOIN STRATEGY COMPARISON Table 2 and Figure 6 compare joining strategies on the synthetic evaluation framework (15

**After:** over 15

### ***Change 261: replace | old pp. [5] | new pp. [5]***

**Before (including deletions):** seeds, mean F1 over

**After:** seeds at each

### ***Change 262: replace | old pp. [5] | new pp. [5]***

**Before (including deletions):** levels L1–L4). The completion-based strategies are the most robust, maintaining F1  $\geq 0.94$  at the highest error severity (L4), compared to F1 = 0.36 for the legacy radius union-find baseline, a 2.6 $\times$  TABLE 2. Join Strategy Leaderboard (Synthetic Eval)

**After:** level.

### ***Change 263: replace | old pp. [5] | new pp. [5]***

**Before (including deletions):** improvement. The radius join's transitive grabbing causes catastrophic over-merging under noise: at L4, it shorts multiple components onto shared nets. We emphasize, however, that

**After:** TABLE 2. Scale-relative completion join on eight of

### ***Change 264: replace | old pp. [5] | new pp. [5]***

**Before (including deletions):** endpoint-graph base joins are themselves far more robust than the radius join, reaching F1 = 0.90

**After:** 15 synthetic circuits

### ***Change 265: replace | old pp. [5] | new pp. [5]***

**Before (including deletions):** (Table 2).

**After:** (16 seeds). Circuit Comps F1 Prec Sim. acc. Parallel R–R 3 1.00 1.00 25% R–R divider 3 0.99 1.00 69% RL series 3 0.99 1.00 81% Diode–R loop 3 0.99 1.00 69% Dual R loops 4 0.93 0.90 12% 4-component loop 4 0.94 0.91 56% Wheatstone bridge 6 0.82 0.85 6% 6-component ring 6 0.95 0.92 50% Besides component-pair F1, we report simulation accuracy: the percentage of seeds for which every source current of the predicted netlist, simulated with the authored values, matches the clean reference within 5%. Table 1 and Figure 5 compare the join strategies. At L4, the scale-relative graph with completion scores F1 0.95, the rescue graph with completion 0.94, the rescue graph alone 0.90, the scale-relative graph alone 0.85 and radius union-find 0.36.

### ***Change 266: insert | old pp. [] | new pp. [5]***

**Before (including deletions):** (none)

**After:** rescue graph differs from the scale-relative graph by a 12 px end-extension and a dead-end rescue step that gives dangling wire ends a longer directional reach. Table 2 breaks down our join by circuit at L4. Precision is at least 0.85 on every

circuit. The Wheatstone bridge is the hardest case, with F1 0.82 and simulation accuracy 6%; the parallel R–R circuit keeps F1 1.00 at 25% simulation accuracy, so the two measures capture different properties. B. WIRE DETECTION Table 3 and Figure 6 summarize the best of 36 extractor variants on the 134-image wire benchmark. Sauvola binarization with a 16 px anchor filter reaches F1 0.976 (median per-image F1 1.000; 87% of images at or above 0.90). Otsu thresholding reaches 0.789 and adaptive Gaussian thresholding 0.845. The configuration in the table deduplicates overlapping segments at 10/18 px; the deployed 12/8 px setting scores 0.973 with the same anchor filter. The score does not depend on image modality: F1 is 0.967 on the 89 photographs and 0.986 on the 45 stroke maps. C. HUMAN-VERIFIED BENCHMARK Table 4 reports every join configuration and baseline on the 31-image human-verified benchmark, together with the paired tests, and Fig. 7 plots the micro-F1 values with their CIs. Our join,

### ***Change 267: replace | old pp. [5] | new pp. [5]***

**Before (including deletions):** adds a further +0.05 over the strongest of these by recovering dropped connections the base graph join leaves floating, while its per-pin edge budget keeps precision high. Pairing that completion with the high-precision scale-relative base (the scale-relative completion join used throughout) gives the best synthetic robustness (0.95 at L4) and, more importantly, the best real-image connectivity (Section IV-D). The large gap is against the legacy radius join; against the strongest prior graph join the improvement is smaller but consistent across every severity level. For readability we omit two intermediate variants whose synthetic scores fall within rounding of rows shown (nearest-pin without anchoring, and

**After:** on

### ***Change 268: delete | old pp. [5] | new pp. []***

**Before (including deletions):** base with end-extension but no dead-end rescue). D. REAL-IMAGE NET-LEVEL EVALUATION The synthetic results above measure robustness under a parametric error model. To validate on real detector output, we constructed, to our knowledge, the first human-verified net-

**After:** (none)

### ***Change 269: replace | old pp. [6] | new pp. [6]***

**Before (including deletions):** 6.

**After:** 5.

### ***Change 270: replace | old pp. [6] | new pp. [6]***

**Before (including deletions):** strategy robustness

**After:** strategies

### ***Change 271: delete | old pp. [6] | new pp. []***

**Before (including deletions):** bars give

**After:** (none)

### ***Change 272: replace | old pp. [6] | new pp. [6]***

**Before (including deletions):** connectivity

**After:** component-pair

### ***Change 273: delete | old pp. [6] | new pp. []***

**Before (including deletions):** The completion-based strategies (left) stay near 1.0, while the radius union-find and nearest-pin baselines (right) collapse at L3–L4.

**After:** (none)

### ***Change 274: replace | old pp. [6] | new pp. [6]***

**Before (including deletions):** connectivity

**After:** component-pair

### ***Change 275: replace | old pp. [6] | new pp. [6]***

**Before (including deletions):** Errors cause fragmentation rather than shorts (precision stays high; Table 4); the balanced Wheatstone bridge is the hardest case. level ground truth for hand-drawn circuits: 31

**After:** TABLE 3. Wire detection on 134

### ***Change 276: replace | old pp. [6] | new pp. [6]***

**Before (including deletions):** from CGHD-1152 in which

**After:** with annotated component boxes. Sauvola rows use 10■/18 px overlap deduplication. Configuration F1 Prec Rec Sauvola + anchor 16 px (ours) 0.976 0.973 0.978 Sauvola + anchor 12 px 0.973 0.974 0.972 Sauvola (w=61) + anchor 12 px 0.959 0.944 0.974 Sauvola (k=0.2875) + anchor 14 px 0.950 0.921 0.980 Otsu threshold + CCL 0.789 0.796 0.783 0.6 0.7 0.8 0.9 1.0 Wire detection F1 (134 images) Triangle (skeleton) Otsu adaptive Gaussian (skeleton) Sauvola (alt.) Sauvola (k=0.2875) Sauvola (w=61) Sauvola, anchor 12px Sauvola, anchor 16px (ours) 0.758 0.789 0.845 0.949 0.950 0.959 0.973 0.976 FIGURE 6. Wire detection F1 across thresholding methods and pipeline variants (134 images, annotated component boxes). The adaptive Gaussian and Triangle bars use skeleton extraction (F1 = 0.845 and 0.758). base, reaches micro-F1 0.890 (P 0.919, R 0.864; macro- F1 0.901; TP/FP/FN 418/37/66; 95% CI [0.855, 0.924]). It beats

### ***Change 277: replace | old pp. [6] | new pp. [6]***

**Before (including deletions):** electrical net was traced and confirmed by a human annotator through a purpose-built verification tool. Because one of our comparison points is a vision-language model (Section IV-F), the human, not any model, is the verifier of record, so the answer key is independent of

**After:** deterministic comparator on

### ***Change 278: replace | old pp. [6] | new pp. [6]***

**Before (including deletions):** method under test. Net-level proposals were bootstrapped by a conservative endpoint-graph join

**After:** test after Holm correction. The margin

### ***Change 279: delete | old pp. [6] | new pp. []***

**Before (including deletions):** dataset's perfect wire labels and then corrected by hand; during verification the human consistently caught cases where the bootstrap join had wrongly split a single rail, and we additionally re-validate all conclusions on the independently authored synthetic ground truth to rule out residual bootstrap bias. Connectivity is scored as component-pair F1 restricted to SPICE-active components, identical to the synthetic metric. Table 3 reports every registered join strategy on the verified set, run on the pipeline's own detected wires. We report micro-averaged (pair-pooled) component-pair F1 as the primary metric, so that a dense 14-component board contributes in proportion to its connections rather than counting the same as a 3-component one; the macro (mean per-image) F1 is listed alongside for reference. The best configuration, the scale-relative completion join (degree-budget completion applied to a high-precision scale-relative endpoint-graph base), achieves micro-F1 = 0.890 (P = 0.92, R = 0.86; 95% bootstrap CI [0.855, 0.924]), improving on the

**After:** (none)

### ***Change 280: replace | old pp. [6] | new pp. [6]***

**Before (including deletions):** default that

**After:** default, which

### ***Change 281: insert | old pp. [] | new pp. [6]***

**Before (including deletions):** (none)

**After:** base, is +0.061 [+0.025, +0.100] with 11 wins, 18 ties and 2 losses. Completion lifts the scale-relative

### ***Change 282: replace | old pp. [6] | new pp. [6]***

**Before (including deletions):** (0.829) and the radius union-find join (0.667) (Figure 7).

**After:** from 0.816 to 0.890, mostly through recall (0.727 to 0.864).

### ***Change 283: replace | old pp. [6] | new pp. [6]***

**Before (including deletions):** gain over that rescue-based completion comes from the cleaner base: the rescue base's 12 px end-extension and dead-end rescue over-reach on noisy detected wires and cost precision, whereas the scale-relative

**After:** scale-relative

### ***Change 284: delete | old pp. [6] | new pp. []***

**Before (including deletions):** keeps precision high while the bounded completion still recovers recall. A reach sweep shows a broad F1 plateau across reach 3–5×s, so the setting

**After:** (none)

### ***Change 285: replace | old pp. [6] | new pp. [6]***

**Before (including deletions):** robust rather

**After:** also more precise

### ***Change 286: delete | old pp. [6] | new pp. []***

**Before (including deletions):** tuned to a spike. Confidence intervals are wide at  $N = 31$  and the per-strategy intervals overlap, but the ranking is consistent and is corroborated on the independently authored synthetic suite (Table 2). TABLE 3. Join Strategy and Baseline Comparison on Human-Verified Net-Level Ground Truth (31 real images, detected wires). Primary metric is micro-averaged (pair-pooled) component-pair F1; macro (mean per-image) F1 is shown for reference. Method F1 Prec Rec F1macro Deterministic joins (ours and ablations) Scale-rel. graph + completion (ours) 0.890 0.919 0.864 0.901 Rescue graph + completion (prior) 0.829 0.789 0.874 0.853 Scale-rel. graph (base) 0.816 0.929 0.727 0.838 Rescue graph (base) 0.787 0.811 0.764 0.813 Radius union-find (legacy) 0.667 0.698 0.638 0.674 Classical deterministic baselines (identical detected wires) Hough + proximity (best) 0.805 0.750 0.868 0.847 Connected-components (best) 0.624 0.965 0.461 0.611 Reference (non-deterministic / oracle wires) VLM (Claude Opus 4.8) 0.923 0.970 0.880 0.949 Scale-rel. + completion, perfect wires 0.890 0.913 0.868 0.916 This real-image ranking resolves an open question from prior work on this pipeline: against bootstrapped pilot labels the degree-budget completion had not consistently beaten its base join; against human-verified ground truth it clearly does (micro-F1 0.829 vs 0.787 for

**After:** (none)

### ***Change 287: replace | old pp. [6] | new pp. [6]***

**Before (including deletions):** alone),

**After:** (0.929 versus 0.811),

### ***Change 288: delete | old pp. [6] | new pp. []***

**Before (including deletions):** the scale-relative variant is better still. The same ordering holds on the authored synthetic suite (Table 2), where the scale- relative

**After:** (none)

### ***Change 289: replace | old pp. [6] | new pp. [6]***

**Before (including deletions):** join also ranks first, so the result is not an artifact

**After:** keeps most

### ***Change 290: replace | old pp. [6] | new pp. [6]***

**Before (including deletions):** how the net-level labels were bootstrapped. The dominant connectivity recipe in prior image-to-netlist systems erases component boxes and reads nets off the connected components, or contours, of the residual wire pixels. It is used by SINA [3], the concurrent contour-based 6 VOLUME 11, 2023

**After:** that precision (0.919, against

### ***Change 291: replace | old pp. [7] | new pp. [6]***

**Before (including deletions):** connectivity

**After:** component-pair

### ***Change 292: delete | old pp. [7] | new pp. []***

**Before (including deletions):** net-level

**After:** (none)

### ***Change 293: replace | old pp. [7] | new pp. [6]***

**Before (including deletions):** (component-pair metric). Our scale-relative completion join leads all deterministic strategies

**After:** with annotated component boxes

### ***Change 294: delete | old pp. [7] | new pp. []***

**Before (including deletions):** the classical Hough and connected-component baselines (run on identical

**After:** (none)

### ***Change 295: replace | old pp. [7] | new pp. [6]***

**Before (including deletions):** wires). The dashed line marks a strong VLM (Claude Opus 4.8) evaluated on the same 31 images, from which our join is statistically indistinguishable (paired micro-F1 difference +0.033, 95% bootstrap CI [-0.009, +0.078]) while being deterministic, simulatable, and orders of magnitude cheaper.

**After:** wires.

### ***Change 296: replace | old pp. [7] | new pp. [6]***

**Before (including deletions):** CIs. node detector

**After:** CIs; the dashed line marks the VLM reference (Claude Opus 4.8). 0.789 for completion on the rescue base). Leaving out any one image moves our micro-F1 within 0.885–0.900, and no deletion changes the sign

### ***Change 297: replace | old pp. [7] | new pp. [6]***

**Before (including deletions):** Peker et al. [5], AMSnet [13],

**After:** any comparison. A sweep of the completion reach over  $\rho \in [3, 5]$  gives macro-F1 0.895–0.903. Several image-to-netlist systems erase component regions

### ***Change 298: insert | old pp. [] | new pp. [6]***

**Before (including deletions):** (none)

**After:** read nets from

### ***Change 299: replace | old pp. [7] | new pp. [6]***

**Before (including deletions):** Bayer instance-segmentation pipeline [6], and was designed for clean printed schematics.

**After:** remaining wire pixels with connected components or contours [1], [2].

### ***Change 300: replace | old pp. [7] | new pp. [6]***

**Before (including deletions):** hand-drawn photographs it has no precise operating point: pencil-stroke fragmentation splits each net into many components ( $F1 < 0.25$  at sensible morphological gap-bridging), and only pathological closing that effectively merges by proximity lifts recall, at a steep precision cost. For a controlled comparison that isolates

**After:** our detected wires,

### ***Change 301: delete | old pp. [7] | new pp. []***

**Before (including deletions):** join algorithm from binarization quality, we run connected- component labeling on our own detected wire segments, identical input to our join, and its

**After:** (none)

### ***Change 302: insert | old pp. [] | new pp. [6]***

**Before (including deletions):** (none)

**After:** connected-component

### ***Change 303: delete | old pp. [7] | new pp. []***

**Before (including deletions):** only

**After:** (none)

***Change 304: delete | old pp. [7] | new pp. []***

**Before (including deletions):** =

**After:** (none)

***Change 305: insert | old pp. [] | new pp. [6]***

**Before (including deletions):** (none)

**After:** at precision 0.965: it rarely merges wrongly, but every stroke gap splits a net

***Change 306: replace | old pp. [7] | new pp. [6]***

**Before (including deletions):** collapses to 0.46: the blobs short or drop most nets), versus 0.890 for the endpoint-graph join (Table 3). A stronger classical alternative, Hough-transform line detection [18] followed by proximity linking, fares better, since

**After:** 0.461). Hough

***Change 307: replace | old pp. [7] | new pp. [6]***

**Before (including deletions):** denoises the fragmented strokes, and

**After:** [20] with proximity linking

***Change 308: delete | old pp. [7] | new pp. []***

**Before (including deletions):** micro-F1 =

**After:** (none)

***Change 309: replace | old pp. [7] | new pp. [6]***

**Before (including deletions):** best-tuned

**After:** best

***Change 310: replace | old pp. [7] | new pp. [6]***

**Before (including deletions):** tolerance, but still trails our join by 0.085 at markedly lower precision. Each baseline is reported at its

**After:** toler- ance. Both baselines use the

***Change 311: replace | old pp. [7] | new pp. [6]***

**Before (including deletions):** over a tolerance sweep, so

**After:** of their sweep on

***Change 312: replace | old pp. [7] | new pp. [6]***

**Before (including deletions):** are upper bounds on the classical recipes. Explicit typed-endpoint reasoning with bounded completion thus outperforms both blob/contour connectivity and line- primitive proximity tracing.

**After:** 31 images, which favors them.

***Change 313: replace | old pp. [7] | new pp. [6]***

**Before (including deletions):** additionally attempted

**After:** also tried

***Change 314: replace | old pp. [7] | new pp. [6]***

**Before (including deletions):** image-to- netlist

**After:** image-to-netlist

### ***Change 315: delete | old pp. [7] | new pp. []***

**Before (including deletions):** end-to-end

**After:** (none)

### ***Change 316: replace | old pp. [7] | new pp. [6]***

**Before (including deletions):** our hand-drawn benchmark, and none could be fairly applied.

**After:** the benchmark.

### ***Change 317: replace | old pp. [7] | new pp. [6]***

**Before (including deletions):** classical printed- schematic

**After:** printed-schematic

### ***Change 318: replace | old pp. [7] | new pp. [6]***

**Before (including deletions):** [2] installs and

**After:** [4]

### ***Change 319: replace | old pp. [7] | new pp. [6, 7]***

**Before (including deletions):** connectivity, a domain- transfer failure.

**After:** con- 6 VOLUME 11, 2023 TABLE 4. Component-pair connectivity on the 31-image human-verified benchmark (annotated component boxes; detected wires unless noted). F1, P and R are micro-averaged. ■ is the micro-F1 of ours minus the comparator with 95% paired bootstrap CI; Holm-adjusted p values are given for the bootstrap, the permutation test on micro-F1 and the Wilcoxon test on per-image F1; W/T/L counts per-image wins, ties and losses from our side. Method F1 P R F1macro Δ[95% CI] Holm p (boot / perm / Wilcoxon) W/T/L Endpoint-graph joins Scale-rel. graph + completion (ours) 0.890 0.919 0.864 0.901 – – – Rescue graph + completion 0.829 0.789 0.874 0.853 +0.061 [+0.025, +0.100] <0.001 / 0.001 / 0.009 11/18/2 Scale-rel. graph (base) 0.816 0.929 0.727 0.838 +0.075 [+0.038, +0.113] <0.001 / 0.002 / 0.005 14/16/1 Rescue graph (base) 0.787 0.811 0.764 0.813 +0.103 [+0.065, +0.142] <0.001 / <0.001 / 0.002 17/14/0 Baselines on the same detected wires Radius union-find (legacy) 0.667 0.698 0.638 0.674 +0.224 [+0.166, +0.281] <0.001 / <0.001 / <0.001 25/6/0 Hough + proximity (best) 0.805 0.750 0.868 0.847 +0.086 [+0.045, +0.127] <0.001 / 0.002 / 0.030 19/8/4 Connected components (best) 0.624 0.965 0.461 0.611 +0.267 [+0.136, +0.409] <0.001 / <0.001 / <0.001 21/8/2 References Ours with annotated wires 0.890 0.913 0.868 0.916 – – – VLM (Claude Opus 4.8) 0.923 0.970 0.880 0.949 –0.033 [–0.078, +0.008] 0.12 / 0.15 / 0.030 7/11/13 nectivity.

### ***Change 320: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** [25]

**After:** [27]

### ***Change 321: delete | old pp. [7] | new pp. []***

**Before (including deletions):** which we ran end-to-end with the released model,

**After:** (none)

### ***Change 322: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** active

**After:** released

### ***Change 323: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** inductor,

**After:** inductor

### ***Change 324: insert | old pp. [] | new pp. [7]***

**Before (including deletions):** (none)

**After:** in

***Change 325: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** circuits contain

**After:** circuits,

***Change 326: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** appearance, so it cannot represent hand-drawn discrete circuits.

**After:** drawings.

***Change 327: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** [4]

**After:** [3]

***Change 328: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** equivalent in spirit to

**After:** which

***Change 329: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** experiment in Section IV-F, rather than a deterministic algorithm. SINA's [3] anonymized

**After:** reference (Section V-I) covers. The

***Change 330: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** was unextractable

**After:** of SINA [1] could not be extracted,

***Change 331: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** [26]

**After:** [28]

***Change 332: delete | old pp. [7] | new pp. []***

**Before (including deletions):** only Co-lab

**After:** (none)

***Change 333: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** released

**After:** trained

***Change 334: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** This survey is itself informative: no prior system provides a deterministic, hand-drawn-ready connectivity module, so the controlled

**After:** The

***Change 335: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** above, both run

**After:** therefore stand in for these approaches

***Change 336: delete | old pp. [7] | new pp. []***



**Before (including deletions):** our

**After:** (none)

### ***Change 337: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** detected wires, are

**After:** inputs. Replacing

### ***Change 338: delete | old pp. [7] | new pp. []***

**Before (including deletions):** fair points of comparison, and our join exceeds both. Running the best join on the dataset's perfect wire labels rather than

**After:** (none)

### ***Change 339: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** leaves

**After:** with the annotated wires, with boxes held fixed, changes

### ***Change 340: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** unchanged at 0.890, while raising

**After:** from 0.8903 to 0.8898 and

### ***Change 341: delete | old pp. [7] | new pp. []***

**Before (including deletions):** only

**After:** (none)

### ***Change 342: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** 0.916 (Table 3), so

**After:** 0.916. On this benchmark, the extracted wires are as good as the annotated ones for joining. The join also scores alike on both image modalities: micro- F1 is 0.883 on the 15 photographs and 0.895 on the 16 stroke maps. D. INDEPENDENT HELD-OUT BENCHMARK The 31-image benchmark has three weaknesses: it is small, its labels were bootstrapped from a graph join similar to ours, and our configuration was chosen on it. We therefore built a second benchmark whose labels come from the CGHD authors' own annotations and whose images were not used to develop or tune the join. The extractor configuration was selected earlier on

### ***Change 343: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** detector accounts

**After:** benchmark, whose resized copies include 25 of the held-out drawings; excluding those 25 changes none of the results below. Reference construction. CGHD provides binary stroke- segmentation maps and symbol instance polygons

### ***Change 344: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** essentially zero pair- weighted F1. The residual error

**After:** a subset of its images. Extending the dataset's own wire-generation tooling, we subtract the dilated symbol polygons from the stroke map and take the remaining connected components as conductors. A conductor connects every polygon it touches, except text; junction polygons merge the conductors they touch. At a crossover polygon, the arms are paired straight through by their local direction ( $180^\circ \pm 45^\circ$ ), and a crossover that cannot be paired

### ***Change 345: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** intrinsic join ambiguity on hand-drawn geometry, not wire detection, consistent

**After:** flagged as ambiguous. Composite out- lines (transformers, optocouplers and relays that contain sub- parts) group their parts but take no contacts. Of 257 images, 179 pass all quality checks; the others have an ambiguous crossover (29), an isolated part (17) or an over-connected part (28). Validation. On the 17 images shared

### ***Change 346: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** 0.976 wire-detection F1 reported above (Table 1).

**After:** human- verified benchmark, the reference agrees with the human nets at micro-F1 0.939 (P 0.993, R 0.891; exact on 11 of 17; 0.964 on the 15 clean images). We inspected every disagreement. Most are pairs the reference misses because of stroke gaps or ambiguous crossings, and seven follow from a convention: the human nets bridge switches. In five pairs, the reference was right and the human labels wrong.

### ***Change 347: insert | old pp. [] | new pp. [7]***

**Before (including deletions):** (none)

**After:** reference mainly under-connects, so it understates the precision of every method. We fixed the quality filter and a junction-handling rule after inspecting these 17 overlaps. Protocol. The held-out set contains the 164 clean images outside the 31, with 3874 reference pairs from 24 drafters (all except drafter\_0, which the detector's training also excluded). The input is the CGHD photograph with its aspect ratio preserved and its long side resized to 1024 px. We chose this resolution on wire-level labels of the 52 wire-benchmark overlaps, before computing any

### ***Change 348: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** degrades

**After:** score. Unlike the 31-image benchmark, which kept

### ***Change 349: delete | old pp. [7] | new pp. []***

**Before (including deletions):** gently with circuit size, with per-image F1 falling from 0.94 on

**After:** (none)

### ***Change 350: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** with  $\leq 6$  components to 0.86 above that and only weak correlation  $r = -0.19$ , and remains high-precision throughout, in contrast

**After:** whose parts all belong

### ***Change 351: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** VLM, whose precision collapses on

**After:** scored device types,

### ***Change 352: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** densest circuits (Section IV-F). E. PER-CIRCUIT ANALYSIS

**After:** held-out set also contains switches, potentiometers, logic gates, speakers, thyristors and transformers (94 of the 164 images contain at least one); these parts take part in the topology but are not scored. Results.

### ***Change 353: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** 4 shows

**After:** 5 gives

### ***Change 354: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** scale-relative completion join's performance across eight representative circuits from the 15-circuit synthetic corpus at L4 (extreme error injection, 16 seeds). TABLE 4. Scale-Relative Completion Join Performance per Circuit at L4 (16 seeds) Circuit Comps

**After:** results. Our join reaches micro-

### ***Change 355: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** Prec Sim. acc. Parallel R-R 3 1.00 1.00 25% R-R divider 3 0.99 1.00 69% RL series 3 0.99 1.00 81% Diode-R loop 3 0.99 1.00 69% Dual

**After:** 0.711 [0.671, 0.748] (P 0.766,

### ***Change 356: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** loops 4 0.93 0.90 12% 4-component loop 4 0.94 0.91 56% Wheatstone bridge 6 0.82 0.85 6% 6-component ring 6 0.95 0.92 50% Precision remains high ( $\geq 0.85$ ) across all circuits even at L4: errors cause fragmentation (dropped connections) rather than shorts (false merges). The Wheatstone bridge is the hard-est topology ( $F1 = 0.82$ ): its balanced arms make the inferred node structure sensitive to a single mis-placed connection,

**After:** 0.664; macro-F1 0.724)

### ***Change 357: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** simulation accuracy falls to 6%. F. VLM CONNECTIVITY EXPERIMENT To quantify the separation argued in Section II, that connectivity is better solved by an explicit geometric model than left to a VLM, we ran a first-party experiment with a strong general VLM (Claude Opus 4.8). To make the comparison fair, the VLM is given exactly what the join receives: the scanned image

**After:** beats every comparator,

### ***Change 358: replace | old pp. [7] | new pp. [7]***

**Before (including deletions):** ground-truth component boxes

**After:** ordering of the endpoint-graph variants as on the human-verified set. Hough linking keeps recall high (0.895) but merges many nets (precision 0.328); the best Hough configuration tuned on this set itself reaches 0.509. The result holds under three changes to the evaluation set: excluding every other photograph of the 31 benchmark drawings (155 images) gives 0.707, and scoring all 257 images including the unclean ones gives 0.626. Across the 24 drafters, our micro-F1 ranges from 0.465 to 0.917

### ***Change 359: replace | old pp. [8] | new pp. [8]***

**Before (including deletions):** labelled

**After:** TABLE 5. Independent held-out benchmark (164 CGHD photographs, 24 drafters, 3874 reference pairs; annotated component boxes). ■ is ours minus the comparator in micro-F1, with 95% paired bootstrap CI. Every difference has Holm-adjusted  $p = 0.0006$ , the resolution floor of the tests. Method F1 P R  $\Delta$ [95% CI] W/T/L Ours 0.711 0.766 0.664 – – Rescue + compl. 0.659 0.622 0.699 +0.053 [0.033, 0.074] 85/44/35 Scale-rel. base 0.595 0.785 0.479 +0.117 [0.097, 0.138] 112/46/6 Rescue base 0.592 0.631 0.558 +0.119 [0.097, 0.142] 113/31/20 Radius u-f 0.518 0.478 0.565 +0.194 [0.159, 0.229] 136/13/15 Hough 0.480 0.328 0.895 +0.231 [0.168, 0.282] 90/22/52 Conn. comp. 0.594 0.798 0.473 +0.117 [0.083, 0.156] 117/12/35 (median 0.709, interquartile range 0.658–0.770), and our join is the best of the evaluated methods for 15 drafters. Scoring conventions. The primary scoring follows the human-verified benchmark: it scores only the electrical device types, treats switches as components that are not scored, and lets separate ground symbols join nodes only through drawn wires. We rescored the same predictions under three alternatives, applied symmetrically to reference and predictions. Adding switches, potentiometers, photoresistors, fuses, lamps, varistors, crystals, speakers and microphones to the scored set (577 more reference pairs in 76 images) gives 0.709, with a margin of +0.055 [+0.036, +0.077] over the rescue-graph completion. Treating every switch as closed gives 0.712 (+0.052). Merging all ground symbols into one node, and all negative-supply symbols into another, adds 778 pairs in 43 images and gives 0.679 (+0.035 [+0.012, +0.059]); the merged ground net concentrates many pairs, so one missed ground attachment costs several. With all three changes combined, our join scores 0.676 and leads every baseline, with Holm-adjusted  $p \leq 0.009$ . Human audit. A human annotator checked the derived nets of 22 held-out images, drawn at random from the 164, in the verification tool used for the human-verified benchmark, correcting any errors. Against the corrected nets, the reference scores micro-F1 0.998 (precision 1.000, recall 0.996; 283 pairs): it was exact on 21 images and missed one pair. Our join scores 0.803 against the corrected nets and 0.801 against the reference on these images, and it leads the rescue-graph completion

### ***Change 360: replace | old pp. [8] | new pp. [8]***

**Before (including deletions):** index,

**After:** +0.054 [+0.018, +0.087], Hough linking by +0.101

### ***Change 361: insert | old pp. [ ] | new pp. [8]***

**Before (including deletions):** (none)

**After:** connected components by +0.196. The annotator verified proposed nets rather than labeling from scratch, so the audit measures how many corrections the reference needs; checking one image took a median of 77 s. Gap to the human-verified

benchmark. The drop from 0.890 to 0.711 comes mainly from the input images. The held-out photographs belong to the CGHD subset for which the dataset provides stroke maps, and for this subset the 704×704 benchmark copies are mostly those stroke maps (16 of the 17 overlaps). Many of these photographs show grid paper or cluttered backgrounds, which the 31-image copies rarely contain; on them, wire F1 falls to about 0.80–0.82, against 0.98 on the benchmark copies. On the 17 overlapping images, our join scores 0.896 on the benchmark copies and 0.728 on TABLE 6. Micro-F1 by number of electrical components per image. The VLM was run on the human-verified benchmark only. Components Images Ours Rescue + compl. Hough VLM Human-verified benchmark (31 images)  $\leq 5$  15 0.940 0.933 0.892 0.982 6–9 4 0.846 0.817 0.843 0.902  $\geq 10$  12 0.886 0.808 0.779 0.912 Held-out benchmark (164 images)  $\leq 5$  41 0.845 0.856 0.835 – 6–9 41 0.779 0.756 0.701 – 10–19 60 0.694 0.656 0.593 –  $\geq 20$  22 0.693 0.608 0.347 – the photographs at long side 1024. Re-weighting the held-out images to the circuit-size distribution of the 31 raises our score by about 0.01, and scoring against the human nets instead of the reference changes it by about 0.01. Input quality explains much of the rest, but not all of it: with the CGHD stroke maps as input instead of the photographs, our held-out score rises only to 0.775. The residual

## Change 362: replace | old pp. [8] | new pp. [8]

**Before (including deletions):** asked

**After:** consistent with some optimism in the 31-image score, on which our configuration was selected. Across both benchmarks, our join ranks first and the endpoint-graph variants keep their order; the classical baselines do not keep theirs (Hough ranks below connected components on the held-out set). E. ROBUSTNESS 1) Circuit complexity Table 6 splits both benchmarks by the number of electrical components. On the 31 images, F1 falls with complexity for every method, including the VLM: across images, our F1 correlates with component count at Spearman  $\rho = -0.53$  ( $p = 0.002$ ). Our recall drop from the  $\leq 5$

## Change 363: replace | old pp. [8] | new pp. [8, 9, 10]

**Before (including deletions):** return,

**After:** the  $\geq 10$  stratum,  $-0.068 [-0.18, +0.05]$  (permutation  $p = 0.32$ ), is not significant. On images with at least ten components, our join beats the rescue-graph completion, the scale-relative base and Hough by  $+0.078$  to  $+0.106$ , with every CI excluding zero, and trails the VLM by  $-0.026 [-0.087, +0.030]$ . On the held-out benchmark, micro-F1 falls from 0.845 for at most five components to 0.693 for 20 or more, while Hough falls to 0.347. 2) Component-size variation within an image A single scale  $s$  per image could fail when component sizes vary within a drawing. On the 31 images, the overall dispersion of component sizes is unrelated to F1 ( $\rho = +0.18$ ,  $p = 0.33$ ). The ratio of the largest to the smallest electrical-component diagonal correlates with F1 at  $\rho = -0.60$ , but it also grows with component count ( $\rho = 0.62$ ); controlling for count, the partial correlation is  $-0.41$  ( $p = 0.024$ ) for our join and  $-0.34$  for the VLM. The size of the largest component relative to  $s$  has partial correlation near zero with F1. The three images that combine an IC with small discrete parts score 0.854 against 0.898 on the other 28. 8 VOLUME 11, 2023 0.0 0.2 0.4 0.6 0.8 1.0 Connectivity micro-F1 (a) Join only: annotated wires 0.35 0.5 0.75 1 1.5 2 3 Image scale factor  $f$  (log scale; native = 1) 0.0 0.2 0.4 0.6 0.8 1.0 Connectivity micro-F1 (b) Full: wires re-extracted 95% bootstrap CI Scale-rel. graph + completion Fixed-px graph + completion Scale-rel. graph (base) Radius union-find (30 px) FIGURE 8. Component-pair micro-F1 on the 31 images resampled by factor  $f$  (native  $f = 1$ ). (a) Join only, with annotated wires rescaled with the image. (b) Full pipeline, with wires re-extracted from each rescaled image. Components are annotated boxes in both panels; shaded bands are 95% bootstrap CIs for our join. 3) Controlled rescaling We resampled all 31 images by factors from 0.35 to 3 and re-projected the annotated boxes and wires (Fig. 8). With annotated wires, our join stays within  $\pm 0.005$  of its native score from  $0.5\times$  to  $3\times$ , with every CI including zero. It drops only at  $0.35\times$  ( $-0.035 [-0.061, -0.005]$ ), where the lower pixel clamps bind for every image. With annotated wires, radius union-find with its fixed 30 px radius falls from 0.690 at native scale to 0.429 at  $2\times$  and 0.153 at  $3\times$ . With wires re-extracted from each rescaled image, the full pipeline scores 0.580, 0.690, 0.811, 0.890, 0.877, 0.866 and 0.866 at 0.35, 0.5, 0.75, 1, 1.5, 2 and  $3\times$ . Upscaling costs at most 0.024 (CIs include zero). The loss under downscaling comes from wire extraction, whose Sauvola window, minimum component area and occlusion margins are set in pixels: at  $0.35\times$  the extractor finds 241 of 675 annotated segments. Completion recovers much of that loss; the scale-relative base alone scores 0.21 at  $0.35\times$ . To separate the contribution of the scale-relative tolerances, we ran the same algorithm with fixed pixel tolerances ( $\tau_{\text{pin}}$ ,  $\tau_{\text{join}}$ ,  $\tau_{\text{t}}$  of 30, 14 and 10 px, completion reach 120 px). It stays within 0.016 of our join between  $0.5\times$  and  $2\times$ . The scale-relative version wins only where the fixed values are most mismatched: by  $+0.015$  with annotated wires at  $0.5\times$  and by  $+0.033$  in the full pipeline at  $3\times$ , both with CIs excluding zero. Most of the scale robustness therefore comes from TABLE 7. Edge-rule ablation on the 31-image benchmark, for the base graph and the full pipeline with completion. Base graph Full pipeline Configuration F1 TP/FP/FN F1 TP/FP/FN All edge types (ours) 0.816 352/27/132 0.890 418/37/66 No T-junctions 0.816 352/27/132 0.890 418/37/66 No rail taps 0.816 352/27/132 0.890 418/37/66 No T-junctions, no rail taps 0.816 352/27/132 0.890 418/37/66 No directional preference 0.816 352/27/132 0.890 418/37/66 Fixed-pixel base tolerances 0.820 356/28/128 0.890 418/37/66 TABLE 8. Mechanism ablations on the 31-image benchmark. “Changed” counts images whose predicted pairs differ from the reference arm. Arm TP/FP/FN F1 F1macro Changed Occlusion on (ours) 418/37/66 0.890 0.901 – Occlusion off 103/26/381 0.336 0.294 31/31 Guard on (ours) 418/37/66 0.890 0.901 – Guard off 418/37/66 0.890 0.901 0/31 Witness required 413/37/71 0.884 0.907 6/31 component-relative endpoint assignment and completion; the scalar tolerances  $\tau = k \cdot s$  contribute a smaller share. The clamps are tuned to native resolution:  $k \cdot s$  falls outside a clamp for 39% of images at native scale, and removing the clamps costs 0.023 in the full pipeline at native scale. 4)

Synthetic mixed-size circuits We redrew one component per synthetic circuit at 0.5, 2 or 3 times its size, keeping only geometrically clean variants, and applied the L1–L4 error model. Enlarging one component costs our join about 0.005 in mean F1, and it stays about 0.03 above the fixed-pixel variant. Shrinking one component costs 0.012. The failure case is a Wheatstone bridge with one 40×15 px arm: in all five variants, the scale-relative methods and radius union-find add one spurious pair even on clean wires, while the fixed-pixel graphs stay exact. A component much smaller than the median is thus the weak point of a median-derived tolerance. F. ABLATIONS Table 7 removes individual edge rules from the base graph, with and without completion. Disabling T-junctions, rail taps or directional preference leaves the aggregate base-graph counts unchanged, so these rules give no measurable gain on this benchmark. Fixed-pixel base tolerances raise base micro-F1 from 0.816 to 0.820. With completion, every row ties at TP/FP/FN 418/37/66. The rescaling and mixed-size experiments (Section V-E) test the tolerance design under conditions that this benchmark does not cover. Table 8 tests the remaining mechanisms. Removing only the occlusion step, with cropping and all extractor settings held fixed, exposes component pixels to the extractor, which then returns 466 wire segments instead of 666; micro-F1 falls from 0.890 to 0.336 (P 0.798, R 0.213), and predic- VOLUME 11, 2023 9 TABLE 9. End-to-end joining with detected components on the 31-image benchmark. The lower block replaces annotated boxes with detector output one factor at a time (confidence 0.5); ■ is the change from the previous row. Components F1 Prec Rec TP/FP/FN Annotated boxes 0.890 0.919 0.864 418/37/66 Detector, conf 0.25 0.613 0.575 0.655 317/234/167 Detector, conf 0.35 0.616 0.582 0.655 317/228/167 Detector, conf 0.50 0.627 0.603 0.653 316/208/168 Sequential step F1 Δ TP/FP/FN Drop missed components 0.759 –0.131 329/54/155 Add spurious detections 0.687 –0.072 322/131/162 Use detected geometry 0.644 –0.043 329/208/155 Use detected classes 0.627 –0.017 316/208/168 tions change on all 31 images. The shared-component guard fires on 18 of 31 images and rejects 38 candidate merges. These merges would join two pins of the same component, which the component-pair metric cannot see, so the scores are unchanged. Requiring a wire witness for every completion edge changes six images and lowers micro-F1 to 0.884 while raising macro-F1 to 0.907. The rescaling experiment adds two further arms, fixed-pixel tolerances and unclamped tolerances (Section V-E), and Section V-G decomposes the loss from detected components. G. END-TO-END WITH DETECTED COMPONENTS We replaced the annotated boxes with the outputs of our trained component detector, a YOLO26m-OBb model with 16 merged classes that reaches a mean average precision (mAP@0.5) of 89.0% on its validation split, with its lowest class recall (70.7%) on crossovers. The detector runs on the CGHD original, corrected for its EXIF orientation tag, at its training size (long side 1024 px), and its boxes are mapped into the benchmark frame. We use the deployment confidence threshold of 0.5 without tuning. On the electrical components, detection precision, recall and F1 are 0.886, 0.858 and 0.872, and the same when the class must also match. As a check of the scoring path, feeding the annotated boxes through the same mapping and matching code reproduces 418/37/66. Table 9 gives the results. End-to-end micro-F1 is 0.627 [0.519, 0.731] (macro-F1 0.656), a change of –0.263 [–0.366, –0.170] from annotated boxes; the lower thresh- olds give 0.613 and 0.616. Replacing the annotated boxes one factor at a time attributes –0.131 to missed electrical components, –0.072 to spurious detections, –0.043 to box localization and –0.017 to classification; with a sequential decomposition, the shares depend on the order of the steps. Missed components cost the most because every pair in- volving a missed component becomes a false negative. The detector is sensitive to orientation (its training used rotation augmentation of at most ±10■). On the 14 images whose benchmark copy is re-oriented relative to the CGHD file, end- to-end micro-F1 is 0.491, against 0.739 on the other 17. H. CROSSOVERS Edge types 2, 4 and 5 do not consult crossover labels, so a missed crossover could merge two independent wires. The 31 images contain 13 annotated crossovers in eight images. With annotated boxes, recall on these eight images is 0.834, against 0.884 on the 23 without a crossover (p = 0.51). In C242\_D1\_P1, suppressing the two endpoint–endpoint edges adjacent to crossings changes TP/FP/FN from 27/4/0 to 19/0/8: the edges that create the false pairs also carry true connections. Deleting the annotated crossover boxes before joining changes predictions on 6 of the 8 crossover images and raises micro-F1 by 0.010, so on annotated boxes the crossover labels do not help the join overall. In the end-to-end run, the detector finds all 13 crossovers and labels each

### ***Change 364: replace | old pp. [8] | new pp. [10]***

**Before (including deletions):** JSON,

**After:** a crossover at every tested threshold, so none of

### ***Change 365: insert | old pp. [] | new pp. [10]***

**Before (including deletions):** (none)

**After:** false pairs in Table 9 comes from a missed or misclassified crossover; removing the detected crossovers lowers end-to-end micro- F1 by 0.019. These images were probably in the detector’s training split, so this is not a held-out estimate of crossover recall, and thirteen instances are too few to test the 70.7% validation recall. I. VLM REFERENCE We ran Claude Opus 4.8 as a reference system. It receives the annotated boxes of the electrical components, drawn on the image and numbered without class labels; our join receives the same boxes with their classes, and neither method receives annotated wires. The VLM was run on the human-verified benchmark only. The model returns JSON

### ***Change 366: replace | old pp. [8] | new pp. [10]***

**Before (including deletions):** we

**After:** each image is a separate call with no shared context. We

### ***Change 367: replace | old pp. [8] | new pp. [10]***

**Before (including deletions):** the resulting component-pairs

**After:** its output

### ***Change 368: insert | old pp. [] | new pp. [10]***

**Before (including deletions):** (none)

**After:** component-pair

### ***Change 369: replace | old pp. [8] | new pp. [10]***

**Before (including deletions):** Each

**After:** The VLM reaches micro-F1 0.923 (P 0.970, R 0.880; macro-F1 0.949) and is exact on 21 of 31 images, against 11 for our join. On pooled pairs, our join trails by  $-0.033$  [ $-0.078$ ,  $+0.008$ ], which is not significant (Holm-adjusted bootstrap  $p = 0.12$ , permutation  $p = 0.15$ ; Table 4). Per image, the VLM is better: it wins 13 images, ties 11 and loses 7 (Wilcoxon  $p = 0.030$ ). Its errors follow the same complexity trend as ours (Table 6). The two systems differ in what they require: the VLM needs a call to a proprietary API for every

### ***Change 370: insert | old pp. [] | new pp. [10, 11]***

**Before (including deletions):** (none)

**After:** and returns no geometric trace of its decisions, whereas our join runs locally and deterministically without connectivity training data, and every recovered connection maps to a wire segment or a completion edge. J. PER-DRAFTER RESULTS Table 10 groups the 31-image results by CGHD drafter, using the drafter paths of the published CGHD-1152 file index, which place every benchmark filename under exactly one drafter. The four drafters with at least four images range from micro-F1 0.854 to 0.987. The held-out benchmark gives the spread over all 24 drafters (Section V-D). 10 VOLUME 11, 2023 TABLE 10. Micro-F1 of our join by CGHD drafter on the 31-image benchmark (drafters with at least four images). Drafter Images F1 F1macro TP/FP/FN drafter\_1 7 0.933 0.907 56/6/2 drafter\_2 5 0.854 0.874 85/14/15 drafter\_10 5 0.987 0.995 37/1/0 drafter\_12 4 0.913 0.938 21/0/4 All 31 0.890 0.901 418/37/66 VI. DISCUSSION The endpoint-graph join with completion is the strongest deterministic method on both real benchmarks, and its margins are significant on both, which differ in labels, images and drafters. The held-out benchmark addresses the three main weaknesses of the human-verified set: its labels come from the dataset authors' annotations and not from a join, it covers 24 of the 25 drafters, and its images were not used to develop the join. Completion accounts for most of the gain over the base graphs, and on the high-precision scale-relative base it adds recall while keeping precision above 0.9. The experiments also locate the remaining errors. With annotated components on clean copies, extracted wires are as good as annotated ones. On difficult photographs, wire extraction limits accuracy; with detected components, missed detections do. Three changes follow from these results: an extractor whose pixel parameters follow the component scale, resampling inputs to a nominal component size, and a detector trained with full rotation augmentation. The VLM reference shows that a general model given component boxes can exceed a geometric join per image on this task. The geometric join remains useful where a local, deterministic and inspectable result is required. A. SCOPE AND LIMITATIONS Task and metric. The pipeline recovers structural connectivity. It does not read component values or device models, so its SPICE export is illustrative and uses placeholder values; simulation of real scans is not evaluated. The component-pair metric ignores terminal identity: a perfect score does not certify pin assignments, the exact net partition or the absence of shorts between pins of one component. The completion guard blocks new same-component merges but does not re-pair merges made by the base graph. Connectivity

### ***Change 371: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** by an independent, context-free model call (one per image),

**After:** from drawn wires only: terminals and supply symbols are wire ends,

### ***Change 372: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** no answer leaks between circuits,

**After:** identically labeled supply terminals (e.g., two "+5 V" points)

### ***Change 373: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** human,

**After:** implied source between a pair of input terminals are

***Change 374: insert | old pp. [ ] | new pp. [11]***

**Before (including deletions):** (none)

**After:** interpreted, which would require reading their labels. Table 11 lists

***Change 375: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** model, remains

**After:** device groups: pin templates for transformers, optocouplers and other complex devices exist but are not validated, and

***Change 376: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** verifier

**After:** detector merges several original classes into one label. Data. All images come from one corpus. The human-verified nets were bootstrapped from a conservative endpoint- graph join and corrected by a single annotator, which may leave a bias toward graph-based methods; on the held-out benchmark, with labels from the dataset's own annotations, our join again ranks first. The inspection

***Change 377: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** record. On

**After:** the 17 overlaps found five human-label errors (four in C242\_D1\_P1); we left the released human-verified labels unchanged. Only 22 of the 164 held-out images were human-audited; the reference mainly misses connections, and its quality filter and

***Change 378: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** clean synthetic control (15 authored circuits)

**After:** junction rule were fixed after inspecting

***Change 379: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** VLM scores  $F1 = 0.99$ , confirming it understands the task and symbol set. On all 31 real scans, evaluated against the human-verified net-level ground truth, it reaches micro- $F1 = 0.923$  ( $P = 0.97$ ,  $R = 0.88$ ; macro- $F1 = 0.949$ ).

**After:** 17 overlapping images.

***Change 380: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** VLM thus edges our deterministic join on aggregate (0.923 vs 0.890 micro- $F1$ ), but the gap is not statistically significant: a paired bootstrap over

**After:** benchmark copies of

***Change 381: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** puts the difference at +0.033 with a 95% CI of  $[-0.009, +0.078]$ ,

**After:** are resized, partly re-oriented, and half stroke maps,

***Change 382: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** includes zero. Two further findings qualify the comparison. First, the VLM's accuracy degrades with circuit complexity: it is exact ( $F1 = 1.00$ ) on 21

**After:** makes them easier than raw photographs

***Change 383: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** 31 circuits, including all the small 3–5-component cases, but

**After:** same drawings. The Hough and CCL baselines keep their 31-image settings

### ***Change 384: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** largest boards it systematically omits connections, with recall falling

**After:** held-out photographs. Scale. The join tolerances derive from one median scale per image, and their clamps are tuned

### ***Change 385: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** 0.63– 0.68 on

**After:** native resolution. A component much smaller than

### ***Change 386: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** three densest boards, exactly where

**After:** median can create

### ***Change 387: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** netlist error is most costly; its precision otherwise stays high (0.97 micro), so its failure mode is dropped nets rather than shorts. The macro-F1 (0.949) is correspondingly inflated by the many easy small circuits, which is why we report the pair-weighted micro-F1

**After:** spuri- ous pair,

### ***Change 388: delete | old pp. [8] | new pp. []***

**Before (including deletions):** primary. Second, the cost and form of the output differ from the geometric pipeline: each image consumes on the order of 105 tokens, the result is free-form text rather than a structured netlist, and nothing

**After:** (none)

### ***Change 389: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** model guarantees electrical validity (e.g. that a two-terminal component is

**After:** synthetic Wheatstone case. The extractor's pixel-valued parameters limit accuracy on downscaled inputs. Rescaling tests changes in zoom,

### ***Change 390: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** shorted). Our geometric join is statistically indistinguishable from the VLM on these circuits while being deterministic, two to three orders of magnitude cheaper, structurally guar- anteed against shorts by the self-loop guard,

**After:** in drawing style,

### ***Change 391: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** simulatable by construction. We therefore treat the VLM as an upper reference for what a frontier model can do here, not as a connectivity module to ship: an explicit model reaches the same accuracy and also gives the determinism, low cost, and structural validity a deployed digitizer needs. V. DISCUSSION

**After:** uses annotated components. Detector.

### ***Change 392: delete | old pp. [8] | new pp. []***

**Before (including deletions):** endpoint-graph model resolves ambiguity that radius- based methods cannot: by representing both wire end- points and component pins as nodes with typed edges, it distinguishes between a wire that passes near a component (endpoint–wire-body edge) and one that connects to it (endpoint–pin edge with directional preference). The component-first assignment further disambiguates by using the geometric constraint that wires must reach a component's pin, not just its bounding box. Degree-budget completion addresses the fundamental lim- itation of

**After:** (none)

### ***Change 393: delete | old pp. [8] | new pp. []***

**Before (including deletions):** noise: even a 97.6% accurate wire detector drops ~2% of wire endpoints, and those dropped endpoints are precisely the connections that matter most for simulation accuracy. The b-matching formulation bounds the



damage from mis-detection (each floating pin gets at most one recovery edge), while the self-loop guard prevents the catastrophic short-circuit failures that plague unconstrained approaches. The human-verified net-level evaluation, while novel for hand-drawn circuits, currently covers 31 images; the 95% bootstrap confidence intervals on the per-strategy F1 are correspondingly wide ( $\pm 0.03$ – $0.05$ ) and overlap between adjacent strategies, so while the top ranking is consistent and corroborated by the independently authored synthetic suite, broader annotation is needed to separate the closely ranked configurations. The net-level labels were bootstrapped by a conservative endpoint-graph join over perfect wires and then human-corrected; we mitigate the residual concern that this favors graph-based methods by validating on synthetic ground truth that is authored from scratch, where the same ranking holds. The evaluation draws from a single hand-drawn corpus (CGHD-1152); generalization across other drawing styles, capture conditions, and printed schematics remains to be validated, as does extension to circuits containing devices outside the simulatable set (switches, transformers, etc.), which we exclude from the net-level benchmark. The VLM connectivity comparison uses a single frontier model (Claude Opus 4.8) on the same 31 images; evaluating additional models and prompting strategies is future work. Component detection at 88.5% mean average precision (mAP@0.5) also leaves room for improvement, particularly on crossovers (70.7% recall). The component-detection model

**After:** (none)

### ***Change 394: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** CGHD-1152 after excluding

**After:** CGHD images with an 85/15 random split, and no manifest of that split survives. The 31 benchmark drawings come from drafters eligible for training, and about 26 of them would be expected in the training split, so the end-to-end numbers are likely optimistic for detection. The detector excludes

### ***Change 395: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** diverged markedly

**After:** differs

### ***Change 396: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** other drafters; this exclusion reduces

**After:** rest of

### ***Change 397: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** training set but avoids domain-shift artifacts from an outlier style. Finally,

**After:** corpus, and it is sensitive to orientation. Coverage. The human-verified benchmark contains 13 annotated crossovers, and we did not evaluate dense multi-track buses or multi-layer crossings. The VLM reference uses one model and one prompt, and was run only on

### ***Change 398: insert | old pp. [] | new pp. [11]***

**Before (including deletions):** (none)

**After:** human-verified benchmark. The

### ***Change 399: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** a first-pass approximation of detector noise rather than a

**After:** authored and not

### ***Change 400: delete | old pp. [8] | new pp. []***

**Before (including deletions):** fit; with the new real-image net-level metric available, calibrating it

**After:** (none)

### ***Change 401: delete | old pp. [8] | new pp. []***

**Before (including deletions):** the

**After:** (none)

***Change 402: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** failure distribution is

**After:** errors. Expansion. The derived reference makes

***Change 403: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** natural next step. VI.

**After:** larger human- verified benchmark affordable: its nets can seed the verification tool, where checking an image took a median of 77 s in our audit. We plan to sample new images stratified by component count, drafter, crossing and bus structure and capture conditions, have two annotators correct each image independently with adjudication of disagreements, and extend the evaluation to printed schematics and to corpora other than CGHD. VII.

***Change 404: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** present

**After:** presented

***Change 405: delete | old pp. [8] | new pp. []***

**Before (including deletions):** complete

**After:** (none)

***Change 406: insert | old pp. [] | new pp. [11]***

**Before (including deletions):** (none)

**After:** structural netlist recovery from

***Change 407: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** circuit digitization,

**After:** schematics,

***Change 408: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** occlusion-first

**After:** occlusion- first

***Change 409: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** extractor that yields an endpoint representation and

**After:** extractor,

***Change 410: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** two-stage endpoint-graph join, together with, to our knowledge, the first human-verified net-level connectivity benchmark for hand- drawn circuits. The

**After:** typed

***Change 411: insert | old pp. [] | new pp. [11]***

**Before (including deletions):** (none)

**After:** and degree- budget completion. With annotated components, the join reaches component-pair micro-F1 0.890 on 31 human- verified images and 0.711 on 164 held-out photographs

***Change 412: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** degree-budget completion is most robust under synthetic error injection (F1 = 0.95 at the highest severity vs. 0.36 for the radius base- line), and,

**After:** independently derived reference nets, and it beats every deterministic baseline on both with significant

### ***Change 413: insert | old pp. [] | new pp. [11]***

**Before (including deletions):** (none)

**After:** margins. Its accuracy holds from 0.5× to 3× image scale when wires are given, and it reaches 0.627 end to end

### ***Change 414: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** high-precision scale-relative base, attains connectivity micro-F1 = 0.890 on real images against human-verified ground truth, beating the prior completion default (0.83) and every deterministic baseline, including the connected-component net-tracing recipe used by recent systems (0.62) and a Hough-plus-proximity line tracer (0.81), all on identical detected wires. Running the join on perfect wires does not raise micro-F1 at all, showing the join, not wire detection, is the bottleneck.

**After:** trained detector.

### ***Change 415: delete | old pp. [8] | new pp. []***

**Before (including deletions):** first-party experiment with a frontier

**After:** (none)

### ***Change 416: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** reaches 0.923 on

**After:** given

### ***Change 417: replace | old pp. [8] | new pp. [11]***

**Before (including deletions):** images, a difference from our join that is not statistically significant (paired 95% CI [−0.009, +0.078]), but at two to three orders of magnitude 8

**After:** components scores higher per image,

### ***Change 418: insert | old pp. [] | new pp. [11, 12]***

**Before (including deletions):** (none)

**After:** 11 TABLE 11. Connectivity evaluation, pin construction and export by device group. Device group Evaluated scope Pin construction Export R/C/L/D/Q, voltage sources and ICs Component-pair connectivity on both benchmarks Geometric templates; endpoint clus- tering for discrete and source types, generic layouts for ICs Placeholder values; ICs are skipped because no export model is imple- mented. Switches Scored in the extended held-out vari- ant (Section V-D); not in the primary subset Two pins at the shorter OBB-edge midpoints; long-axis fallback A 0.001 Ωresistor if the two pins map to distinct nodes; no switch state is recovered. Transformers, thyristors, optocouplers and other complex devices Not evaluated Two-pin geometric guess for transformers; optocoupler template; generic fallback No export model; skipped. at the cost of a proprietary call per image. Better extraction on difficult photographs and better component detection are the most direct routes to

### ***Change 419: replace | old pp. [9] | new pp. [12]***

**Before (including deletions):** cost and without simulatable, structurally valid output, supporting the explicit geometric design. Future work in- cludes scaling the human-verified benchmark, calibrating the synthetic error model to the measured detector distribution, and extending coverage to devices outside the simulatable set.

**After:** end-to-end accuracy.

### ***Change 420: replace | old pp. [9] | new pp. [12]***

**Before (including deletions):** IV-F,

**After:** V-I,

### ***Change 421: replace | old pp. [9] | new pp. [12]***

**Before (including deletions):** evaluation scripts,

**After:** the

### ***Change 422: replace | old pp. [9] | new pp. [12]***

**Before (including deletions):** genera- tor,

**After:** generator,

***Change 423: replace | old pp. [9] | new pp. [12]***

**Before (including deletions):** join- strategy

**After:** CGHD-derived reference nets of the held-out bench- mark (ground\_truth/cghd\_ref/), the join-strategy

***Change 424: delete | old pp. [9] | new pp. []***

**Before (including deletions):** connected-component

**After:** (none)

***Change 425: delete | old pp. [9] | new pp. []***

**Before (including deletions):** and

**After:** (none)

***Change 426: insert | old pp. [] | new pp. [12]***

**Before (including deletions):** (none)

**After:** and the evaluation scripts for the paired tests, strata, rescaling, held-out and end-to-end experiments (wire\_detection/benchmark/revision2/)

***Change 427: replace | old pp. [9] | new pp. [12]***

**Before (including deletions):** avail- able

**After:** available

***Change 428: replace | old pp. [9] | new pp. [12]***

**Before (including deletions):** <https://github.com/boscochanam/circuit-digitization>.

**After:** <https://github.com/boscochanam/circuit-digitization>, together with the image-to-drafter mapping used in Ta- ble 10. The reference nets are derived from the CGHD annotations [14]; no CGHD images are redistributed.

***Change 429: replace | old pp. [9] | new pp. [12]***

**Before (including deletions):** the CGHD-1152 dataset [1].

**After:** CGHD images [14].

***Change 430: delete | old pp. [9] | new pp. []***

**Before (including deletions):** F. Thoma, J. Bayer, Y. Li, and A. Dengel, "A public ground-truth dataset for handwritten circuit diagram images," in Proc. Int. Conf. Document Analysis and Recognition (ICDAR), 2021, pp. 20–27. [Online]. Available: [https://doi.org/10.1007/978-3-030-86198-8\\_2](https://doi.org/10.1007/978-3-030-86198-8_2) [2] C. R. Kelly and J. M. Cole, "Digitizing images of electrical-circuit schematics," APL Mach. Learn., vol. 2, no. 1, p. 016109, 2024. [3]

**After:** (none)

***Change 431: replace | old pp. [9] | new pp. [12]***

**Before (including deletions):** [4] J. Bhandari, V. Bhat, Y. He, H. Rahmani, S. Garg, and R. Karri, "Masala- CHAI: A large-scale SPICE netlist dataset for analog circuits by harnessing AI," arXiv:2411.14299, 2024. [5]

**After:** [2]

***Change 432: replace | old pp. [9] | new pp. [12]***

**Before (including deletions):** [6]

**After:** [3] J. Bhandari, V. Bhat, Y. He, H. Rahmani, S. Garg, and R. Karri, "Masala- CHAI: A large-scale SPICE netlist dataset for analog circuits by harnessing AI," arXiv:2411.14299, 2024. [4] C. R. Kelly and J. M. Cole, "Digitizing images of electrical-circuit schematics," APL Mach. Learn., vol. 2, no. 1, p. 016109, 2024. [5]

***Change 433: replace | old pp. [9] | new pp. [12]***

Before (including deletions): [7]

After: [6]

***Change 434: replace | old pp. [9] | new pp. [12]***

Before (including deletions): [8]

After: [7]

***Change 435: replace | old pp. [9] | new pp. [12]***

Before (including deletions): [9]

After: [8]

***Change 436: replace | old pp. [9] | new pp. [12]***

Before (including deletions): [10]

After: [9]

***Change 437: replace | old pp. [9] | new pp. [12]***

Before (including deletions): [11]

After: [10]

***Change 438: replace | old pp. [9] | new pp. [12]***

Before (including deletions): [12]

After: [11]

***Change 439: insert | old pp. [] | new pp. [12]***

Before (including deletions): (none)

**After:** [12] S. Gao, F. Shi, X. Xu, J. Chen, H. H.-C. Lu, et al., "A memristor-based neural network circuit with classical conditioning and fear generalization," IEEE Trans. Consumer Electron., 2026, doi: 10.1109/TCE.2026.3663359.

***Change 440: insert | old pp. [] | new pp. [12]***

Before (including deletions): (none)

**After:** S. Gao, Y. Song, Y. Cao, H. H.-C. Lu, L. Qin, Y. Zhang, and J. Mou, "Biologically plausible memristive decision-making circuit for adaptive control in industrial autonomous navigation," IEEE Trans. Ind. Informat., vol. 22, pp. 4801–4812, 2026, doi: 10.1109/TII.2026.3668794. [14] F. Thoma, J. Bayer, Y. Li, and A. Dengel, "A public ground-truth dataset for handwritten circuit diagram images," in Proc. Int. Conf. Document Analysis and Recognition (ICDAR), 2021, pp. 20–27. [Online]. Available: [https://doi.org/10.1007/978-3-030-86198-8\\_2](https://doi.org/10.1007/978-3-030-86198-8_2) [15]

***Change 441: replace | old pp. [9] | new pp. [12]***

Before (including deletions): [14]

After: [16]

***Change 442: replace | old pp. [9] | new pp. [12]***

Before (including deletions): [15]

After: [17]

***Change 443: replace | old pp. [9] | new pp. [12]***

Before (including deletions): [16]

After: [18]

***Change 444: replace | old pp. [9] | new pp. [12]***

Before (including deletions): [17]

After: [19]

***Change 445: replace | old pp. [9] | new pp. [12]***

Before (including deletions): Kulkarni

After: Kulkarni, V. Kumar, R. L. Minz, M. Varshney, T. Samon, A. Mitra,

***Change 446: replace | old pp. [9] | new pp. [12]***

Before (including deletions): ICCV Workshops,

After: IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW),

***Change 447: replace | old pp. [9] | new pp. [12]***

Before (including deletions): [18]

After: [20]

***Change 448: replace | old pp. [9] | new pp. [12]***

Before (including deletions): [19]

After: [21]

***Change 449: replace | old pp. [9] | new pp. [12]***

Before (including deletions): [20]

After: [22]

***Change 450: replace | old pp. [9] | new pp. [12]***

Before (including deletions): [21]

After: [23]

***Change 451: replace | old pp. [9] | new pp. [12, 13]***

Before (including deletions): [22]

After: 12 VOLUME 11, 2023 [24]

***Change 452: replace | old pp. [9] | new pp. [13]***

Before (including deletions): [23]

After: [25]

***Change 453: replace | old pp. [9] | new pp. [13]***

Before (including deletions): [24]

After: [26]

***Change 454: replace | old pp. [9] | new pp. [13]***

Before (including deletions): [25]

After: [27]

***Change 455: replace | old pp. [9] | new pp. [13]***

Before (including deletions): [26]

After: [28]

***Change 456: delete | old pp. [9] | new pp. []***

**Before (including deletions):** VOLUME 11, 2023 9

**After:** (none)

***Change 457: delete | old pp. [10] | new pp. []***

**Before (including deletions):** 10

**After:** (none)

***Change 458: insert | old pp. [] | new pp. [13]***

**Before (including deletions):** (none)

**After:** 13